FUNCTIONAL SPECIFICATION DOCUMENT

Nexus: Incident Report Management System

DOCUMENT VERSION 4.0

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1/17/2025	2.2	The authors updated the Context Flow Diagram and Data Flow Diagram.	Carlina Chyle Evangelista, Enki A. Prince Alava
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1/30/2025	3.2	The authors accomplished the data migration, conversion requirements and system configurations.	Jeruh A. Fornal, Patrick Jeremie Maniaul, Vince Lloyd Mondares
1/30/2025	4.0	The authors finalized the content and accomplished the documentation of the system.	Joshua C. Africa, Carlina Chyle Evangelista

APPROVALS

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1/02/2025	1.0	Project Manager	Jeruh A. Fornal
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1. Introduction

In the process of handling incident reports in Phase 1 Homeowner's Association at San Lorenzo South Subdivision it usually involves traditional handling of incidents issues by physically reporting the concerns to the grievance committee. Upon receiving the reports, the Chairman of the Grievance and Adjudication Committee of homeowner's association documents the incident through paper forms or spreadsheets that leads to human errors and difficulties when tracking the incident status of each report. Manual managing of incidents by categorization can cause inefficacies and miscommunications. This traditional system lacks centralized data storage, reducing transparency that leads to dissatisfaction of residents. As a result, there is a need for an organized approach to handle incident report management, which the proposed Nexus: An Incident Report Management System aims to address.

1.1 Purpose of the document

This Functional Specification Document serves as a comprehensive guide for residents and committee officials involved in filing, managing, and verifying incident reports within the Incident Report Management System (IRMS). It outlines the objectives and processes to ensure efficient handling of incident records. The Nexus consists of a summary of reports filed by the residents via an online form. All reports can be viewed, updated, archived, or generate summary reports of incidents. In this module also includes the sending of acknowledgement request through email to the residents involved in the incident. By moving from traditional paper-based filing to an online system, this project aims to make incident reporting faster, more organized, and accessible for everyone involved.

1.2 Project Scope

The Incident Report Management System (IRMS) is a management system designed to manage incident reports of residents in Phase 1 San Lorenzo South Subdivision. This aims to digitalize the traditional handling of reports to improve transparency and efficiency in handling incident reports. IRMS allows residents to report an incident online by accessing the site.

By providing the necessary information for incident description and causes, the HOA's Grievance and Adjudication Committee can immediately file the report to the system. This allows residents a quick way to report a concern that needs immediate action. For Admins, it includes a page where the grievance committee can view the list of all incidents that can be managed by sorting the date, status, name, and incident category, and view the details of incidents.

1.3 Scope of the document

The scope of this project is to manage the incident reports in Phase 1 San Lorenzo South digitally. The present document indicates the requirements, functionality, and processes for an incident report management system. It guides developers, administrators, and stakeholders in the appropriate management of incident reports in a community.

The Nexus enables residents to file incidents online by filling out a form with personal information, a valid ID for verification, and a description of the incident, which is then reflected at the HOA's headquarters in the Phase 1 Clubhouse. The system includes features such as an Automated Acknowledgement Email to update complainants once their issue is resolved, a View Reports function to organize and access submitted reports, and an Update Report Status feature for tracking the progress of reports (e.g., pending, declined). Resolved incidents can be archived, and the system also includes a Generate Report feature to summarize the data. An Acknowledgement Request email is sent to residents, informing them of the status of their reports. The Dashboard Module provides real-time statistics on the number of submitted reports, their status, and the performance of the Grievance and Adjudication Committees, with quick access to visualizations like graphs and summary reports for easy tracking.

1.4 Related documents

Component	Name (with link to the document)	Description
Gantt Chart	Gantt Chart https://iskolarngbayan-team-e8duyulh.atlassian.net/wiki/x/c4EB	Gantt chart is a visual project management tool that shows tasks or activities over time, helping track progress, manage timelines, and ensuring proper project planning.
Project Roadmap	Roadmap https://iskolarngbayan-team-e8duyulh.atlassian.net/wiki/x/ogEB	The roadmap visualizes the various phases and objectives of the system, providing the timeframe for its projected progress.
Project Milestone	Project Milestone https://iskolarngbayan-team-e8duyulh.atlassian.net/wiki/x/v4Az	The project milestone is a significant event in a project timeline that includes key stages or achievements, such as the completion of a phase,

		delivery of a major component, or reaching a specific goal, which helps track progress and ensures it stays on schedule.
UI/UX Design Specification	Figma Interface Mockups https://www.figma.com/design/ehivd67sw0NXhSatvNCesi/IRMS_Group%236?node-id=7-73&t=RBmkFZotzC6qNMZR-1	Figma is used for an interface representation including page layouts and user flows.
Project Management Tool.	IRMS – Jira Task List https://iskolarngbayan-team-e8duyulh.atlassian.net/jira/software/projects/NIR/boards/2?atlOrigin=eyJpIjoiMGY4NTM5YjQwNGE4NGQxOGE4MDg5ZDRmOTVhMTk5MmYiLCJwIjoiaiJ9	The project management tool contains a detailed list of tasks that are tracked for the project development, including updates and bugs.

1.5 Terms/Acronyms and Definitions

Term/Acronym	Definition	Description
Incident Report Management System (IRMS)	A process of recoding incidents and concerns to ensure safety and perform immediate action to the issue.	It refers to the organized management records of incident reports such as disputes, violations, and accidents.
Homeowner's Association	It is a small private organization that governs the housing community and sets rules to manage the welfare of its local community.	A homeowner's association is a small village is a communal organization that maintains common places, enforces village laws, and fosters resident peace. It oversees maintaining common spaces, resolving complaints, and organizing community-building events.

Grievance and Adjudication Committee	A committee responsible for addressing complaints, disputes, and conflicts, as well as making formal decisions on incident reports or issues raised by individuals or groups.	This committee ensures fair handling of grievances by investigating issues, conducting hearings, and delivering judgments following established procedures to maintain order and harmony.

1.6 Risks and Assumptions

The risks are an unavoidable component of any system development project and cannot be totally eradicated. Identifying, assessing, and managing risks is crucial for minimizing their effect and chance of recurrence.

Given the assumption that the residents may have access to internet, some potential risks that you may encounter are included below:

- User Accessibility: Majority of residents own gadgets and capable of accessing the system. However, some residents have limited access to the internet due to poor internet connectivity and lack of necessary digital literacy to navigate the system.

- Technical Failures: System may encounter server crashes, databases failure, and downtime that prevents residents from reporting incident reports online.
- User Error: Residents may submit incorrect information due to typos that can result to inaccurate data.

- Fake Incident Reports: Attackers could submit fake reports that could spam and slow down the system, which leads to bad performance of the system.

2. System/ Solution Overview

Nexus is an incident report management system specifically designed for Phase 1 HOA's Grievance and Adjudication Committee of San Lorenzo South Subdivision. It aims to modernize the filing of incidents and provide a faster and convenient way for the community to report concerns to the association. This system is aimed to improve the management of incident records that provides quick access to the part of administrator personnels for transparency and tracking.

2.1 Context Diagram/ Interface Diagram/ Data Flow Diagram, Application Screen Flow, Sitemap, Process Flow

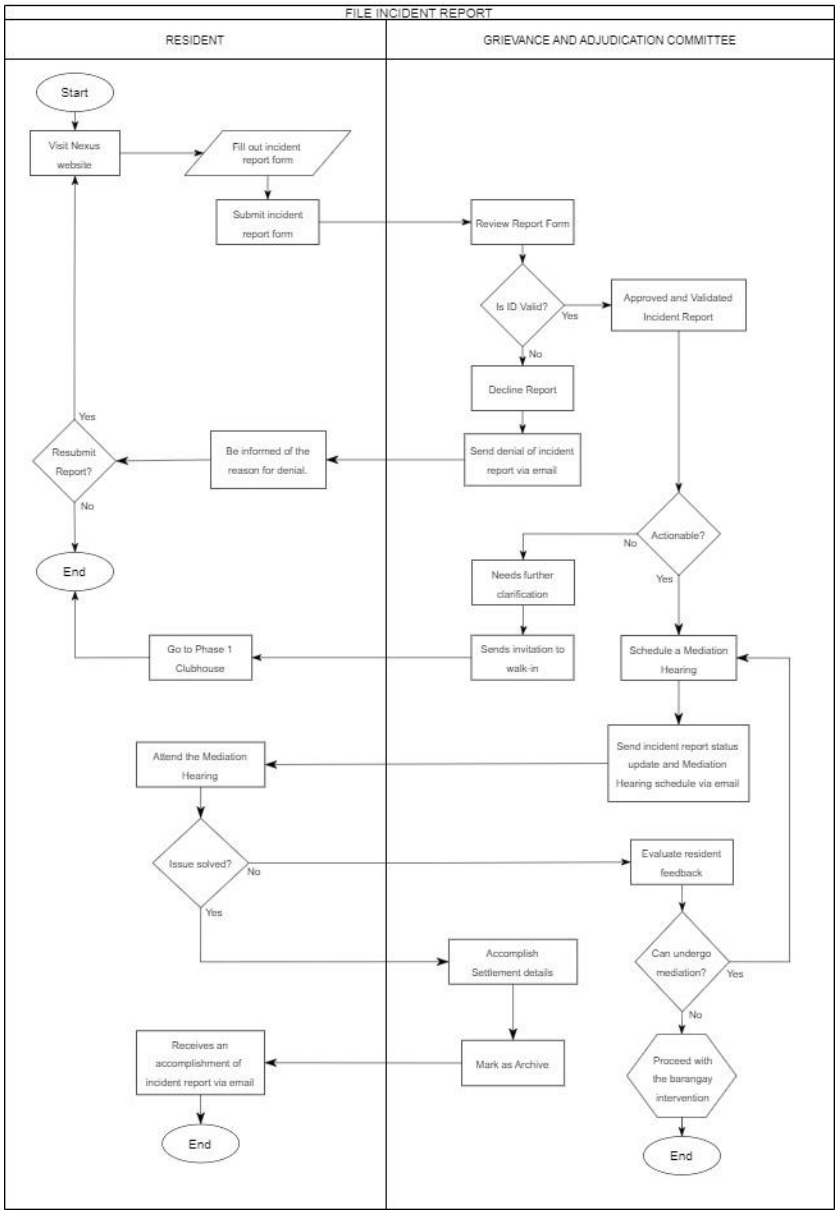


Figure 1. Proposed Business Process Flow Diagram

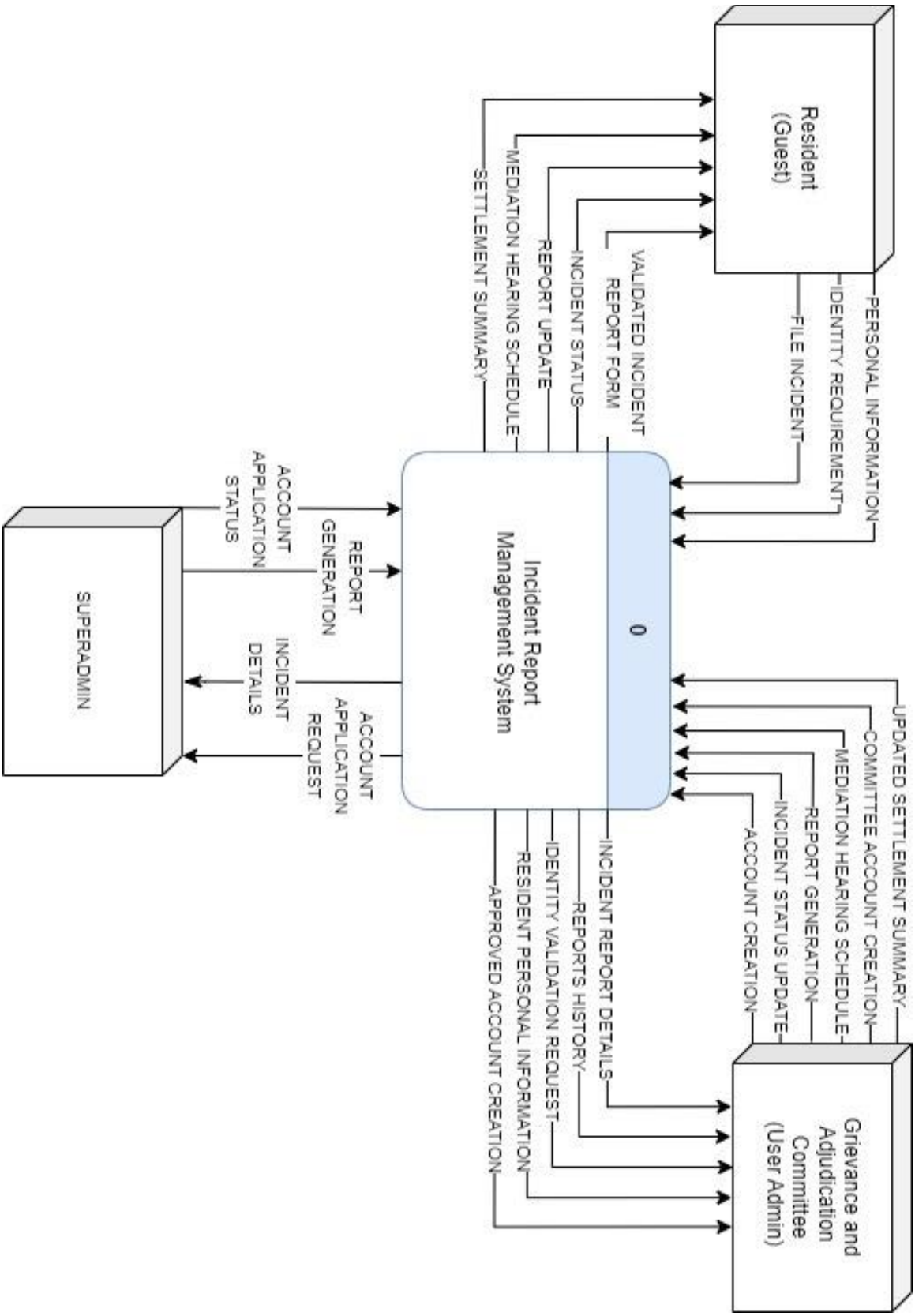


Figure 2. Proposed Context Flow Diagram

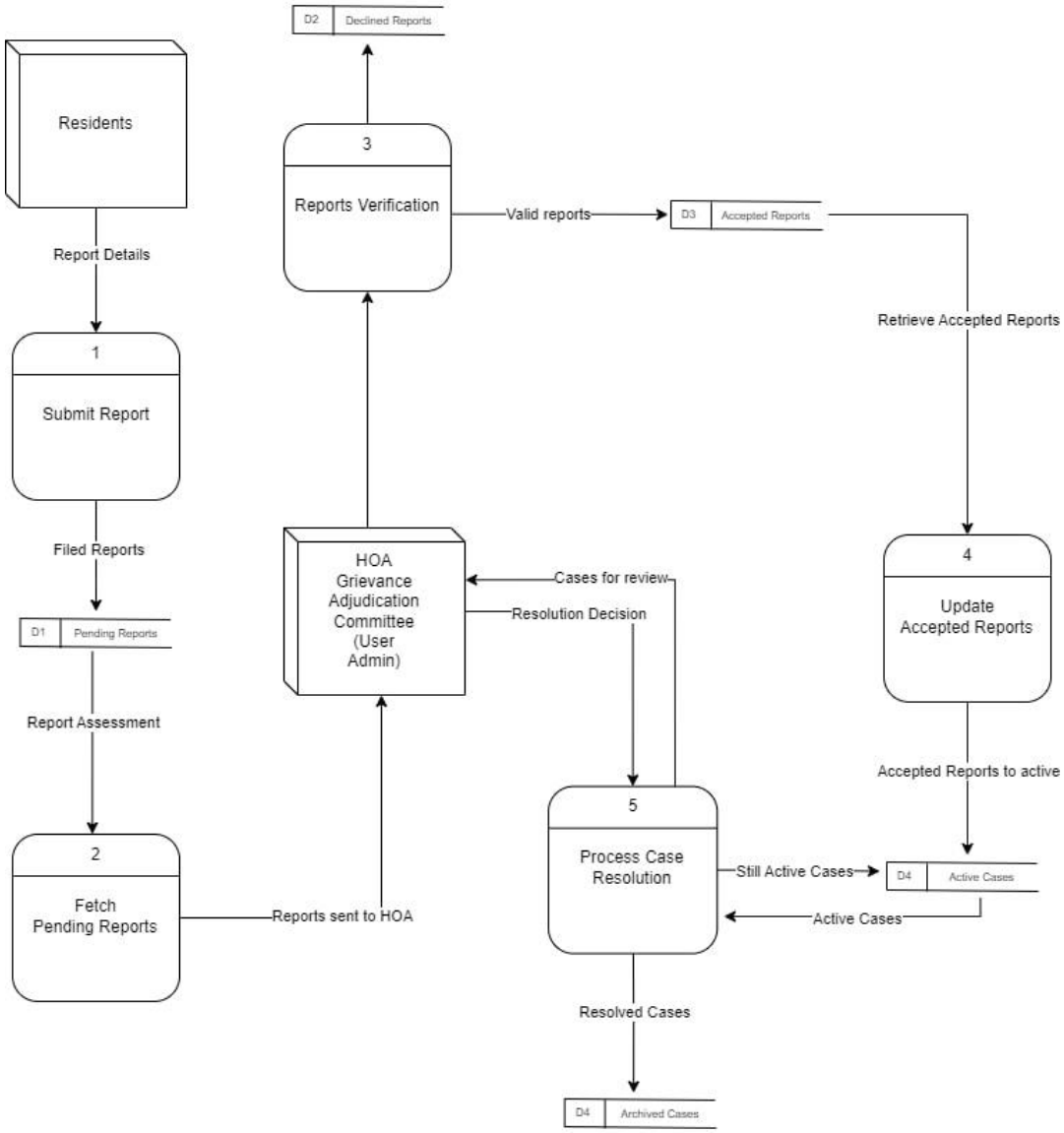


Figure 3. Data Flow Diagram – Level 0 Incident Report Management System

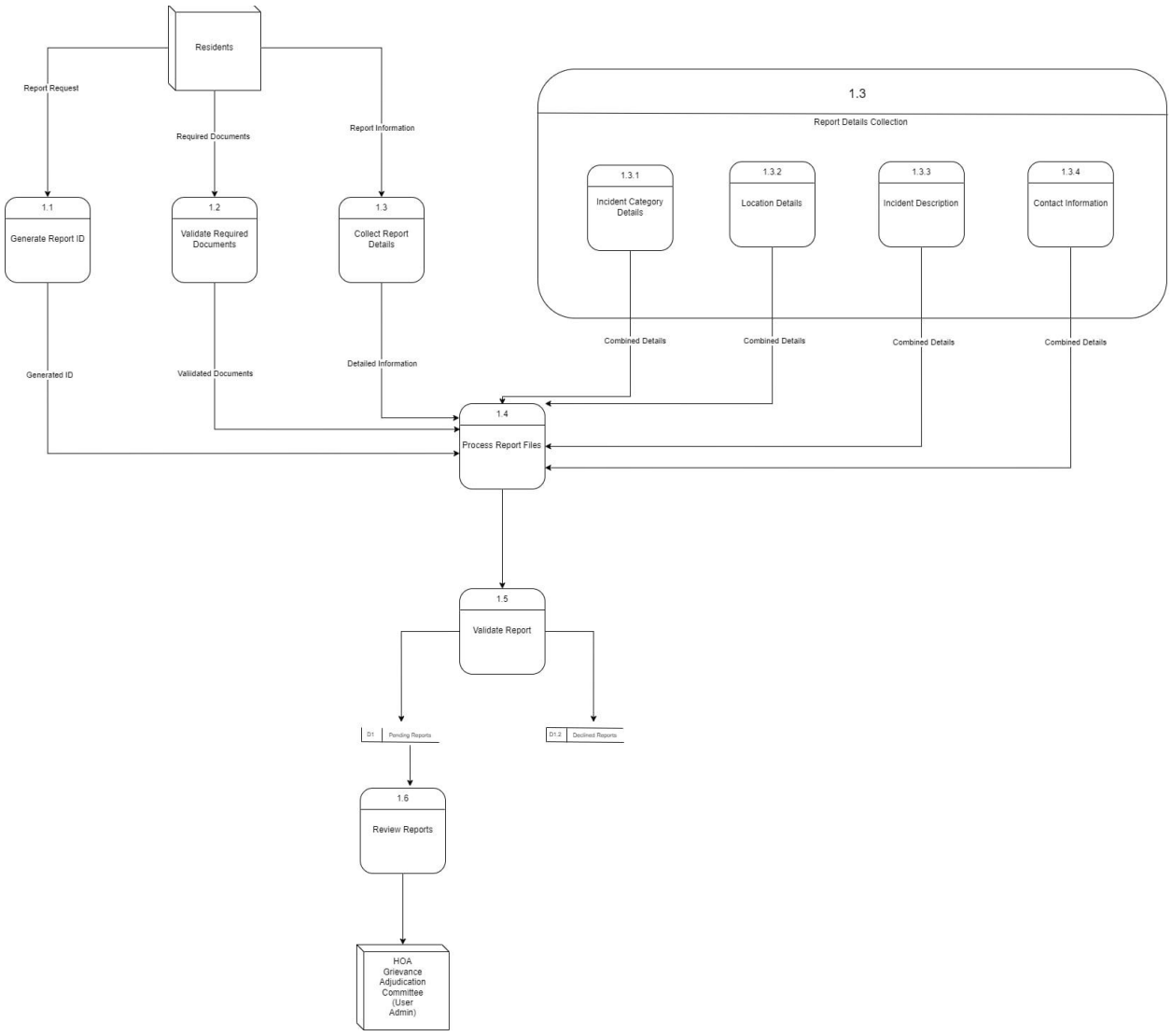


Figure 4. Data Flow Diagram – Level 1 File Incident Report

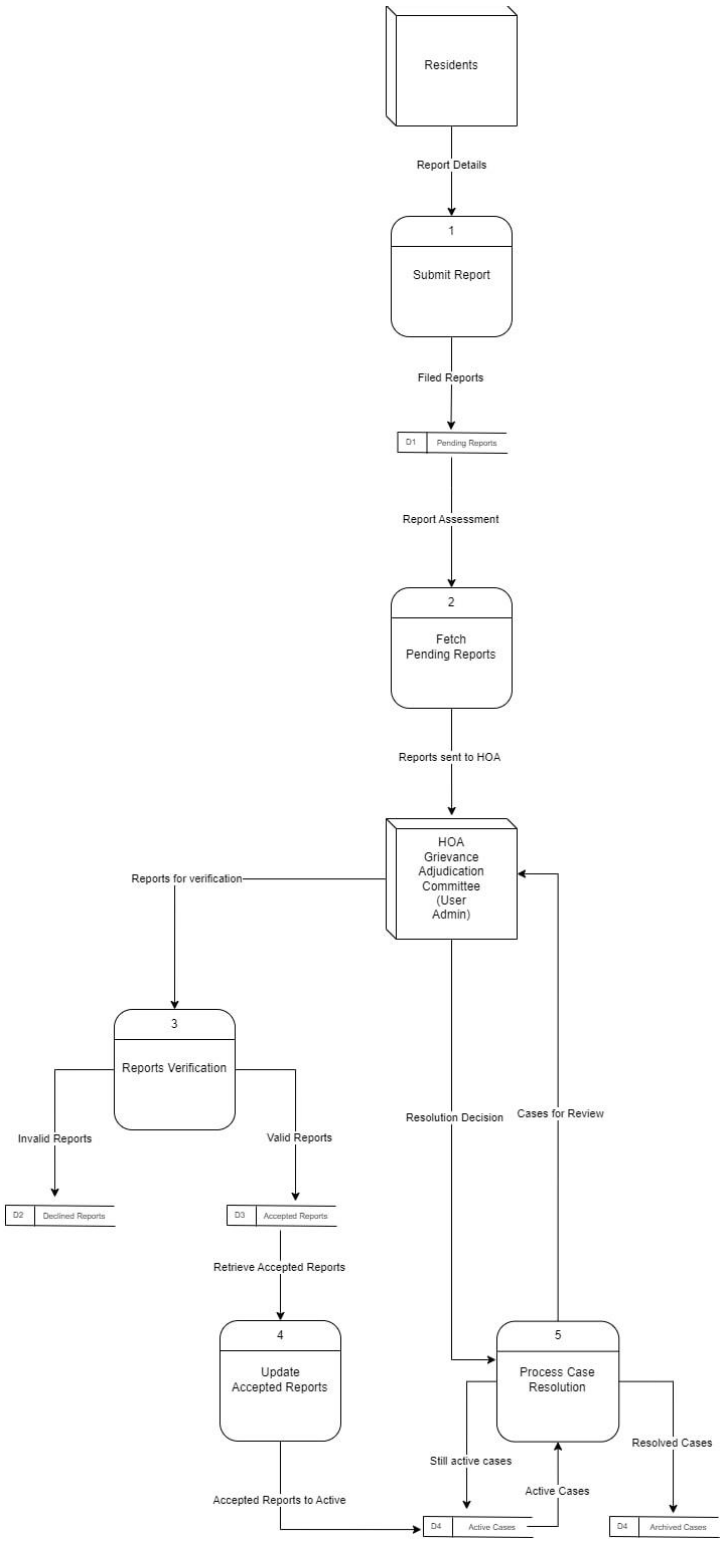


Figure 4.1 Data Flow Diagram – Level 1 Report Management

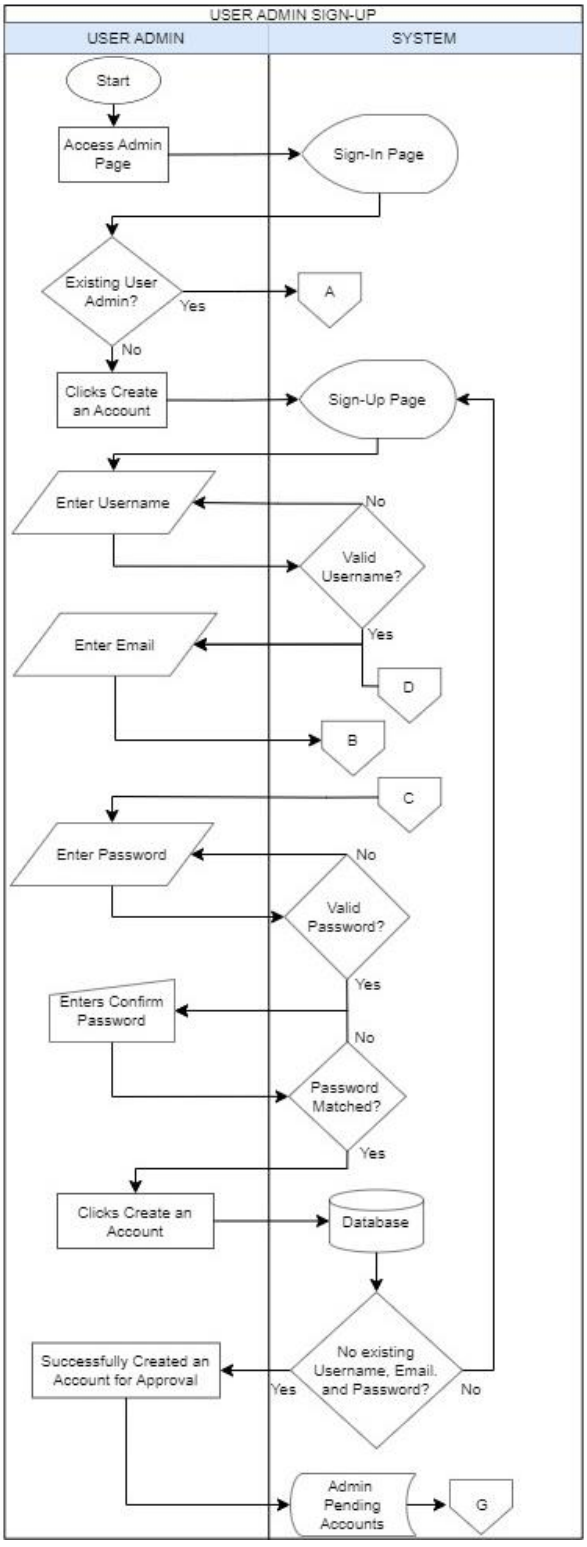


Figure 5. System Flow Diagram – User Admin Sign-Up

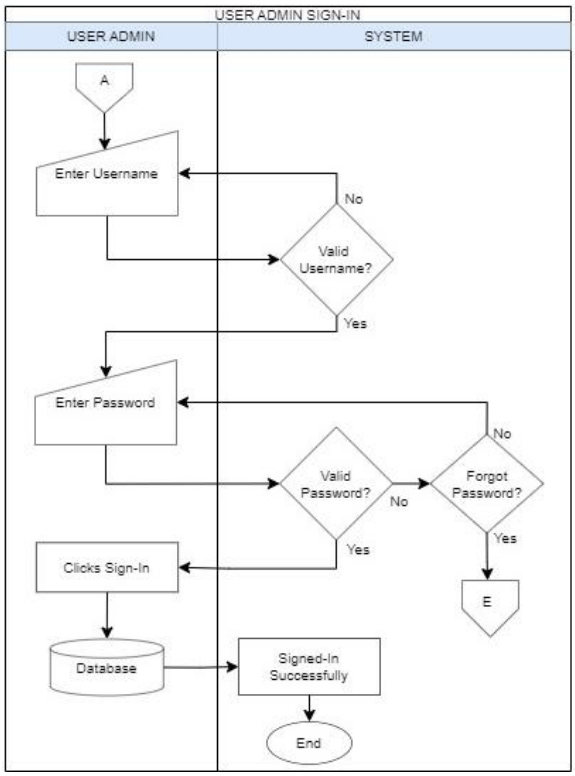


Figure 6. Proposed System Flow Diagram – Admin Sign-In

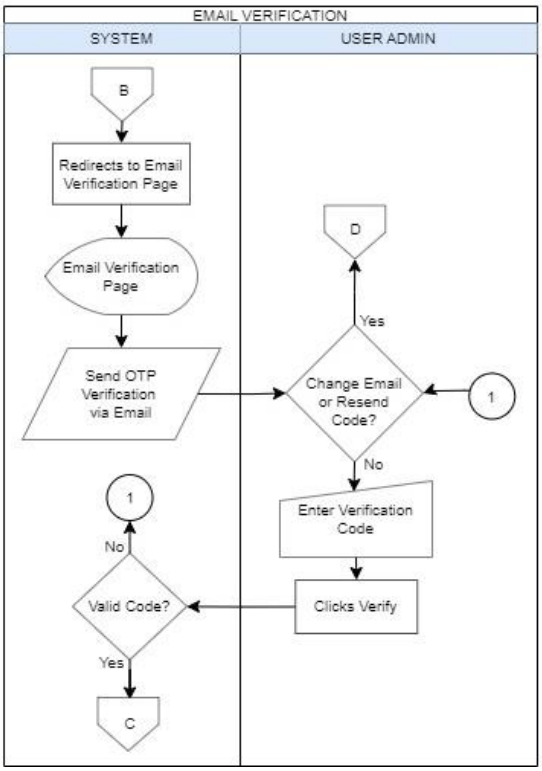


Figure 7. Proposed System Flow Diagram – Email Verification

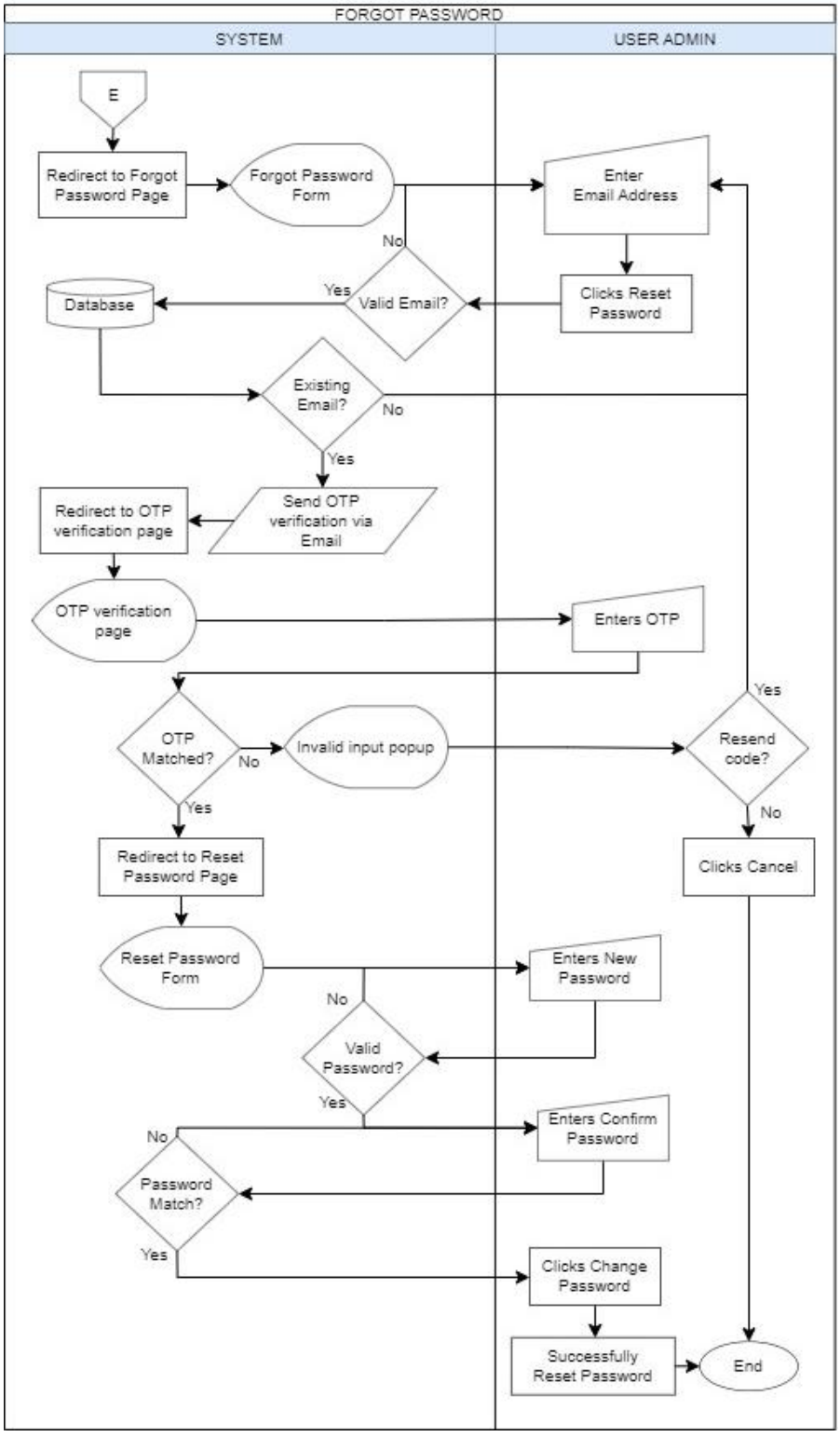


Figure 8. Proposed System Flow Diagram – Forgot Password

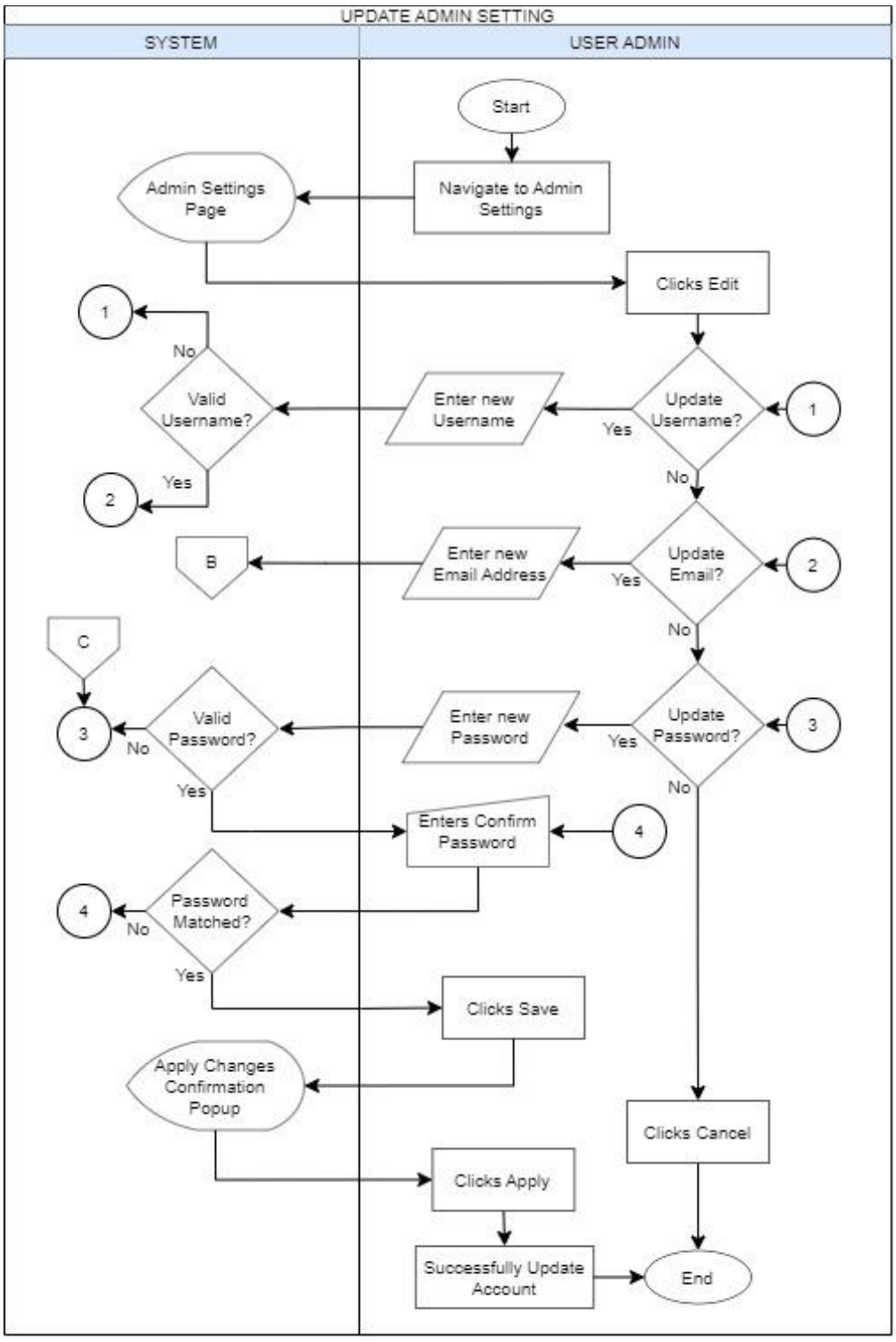


Figure 9. Proposed System Flow Diagram – Update Admin Settings

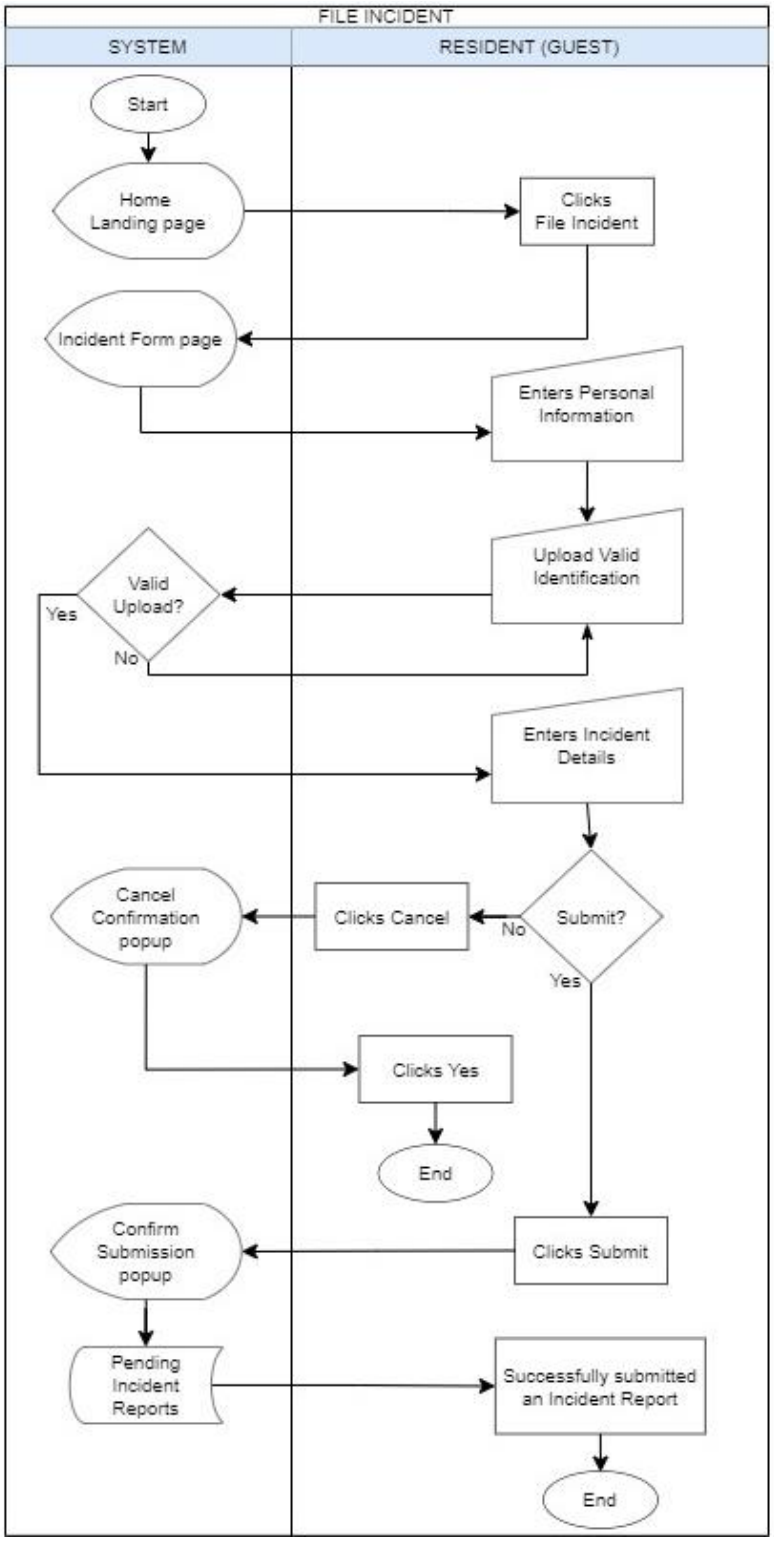


Figure 10. Proposed System Flow Diagram – File Incident

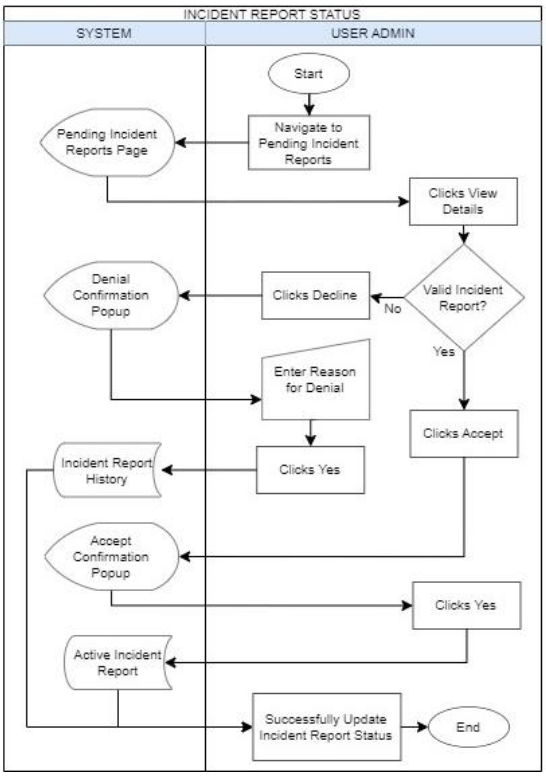


Figure 11. Proposed System Flow Diagram – Incident Status

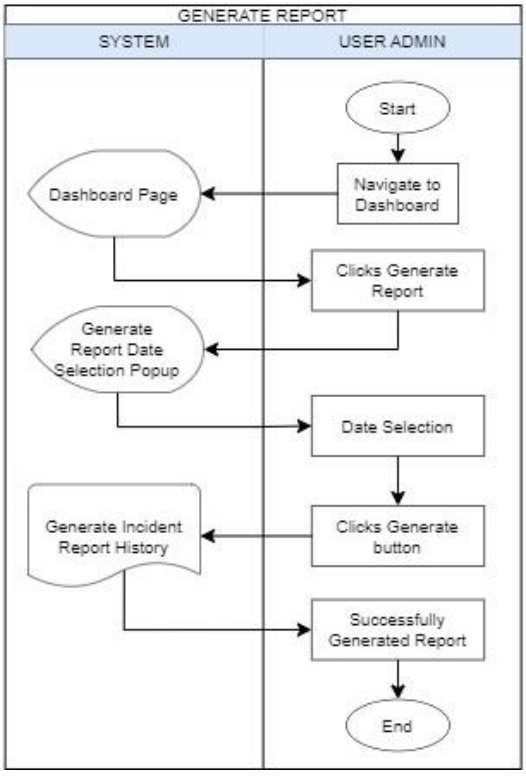


Figure 12. Proposed System Flow Diagram – Generate Report

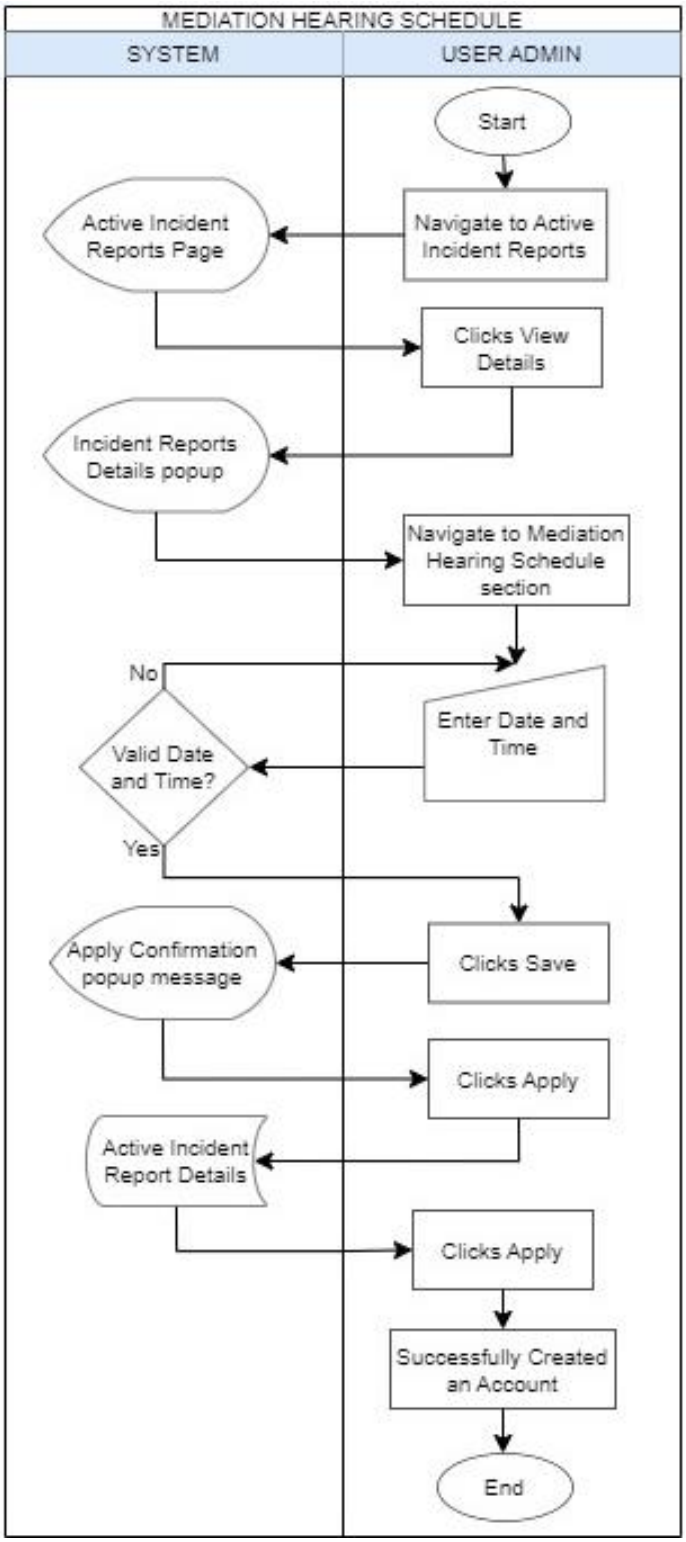


Figure 13. Proposed System Flow Diagram – Mediation Hearing Schedule

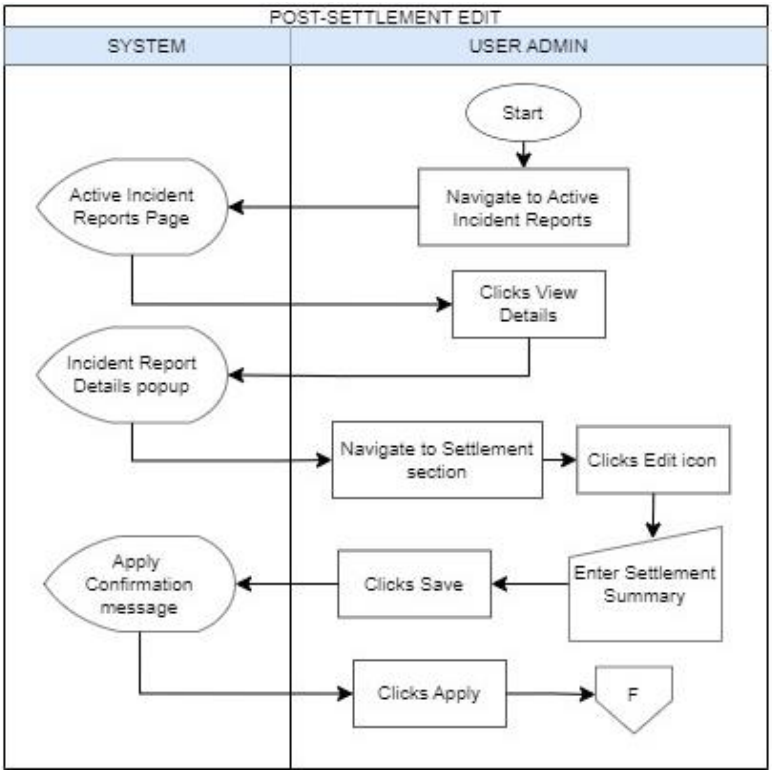


Figure 14. Proposed System Flow Diagram – Post-Settlement Edit

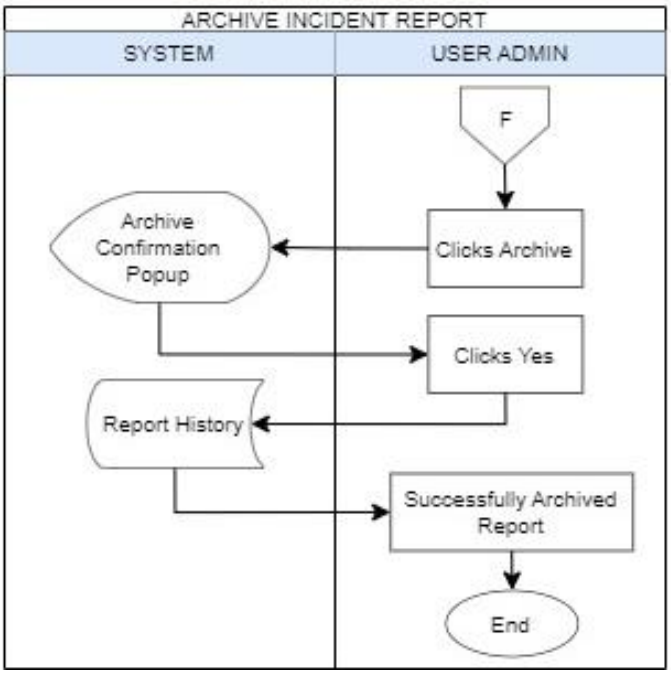


Figure 15. Proposed System Flow Diagram – Archive Incident Report

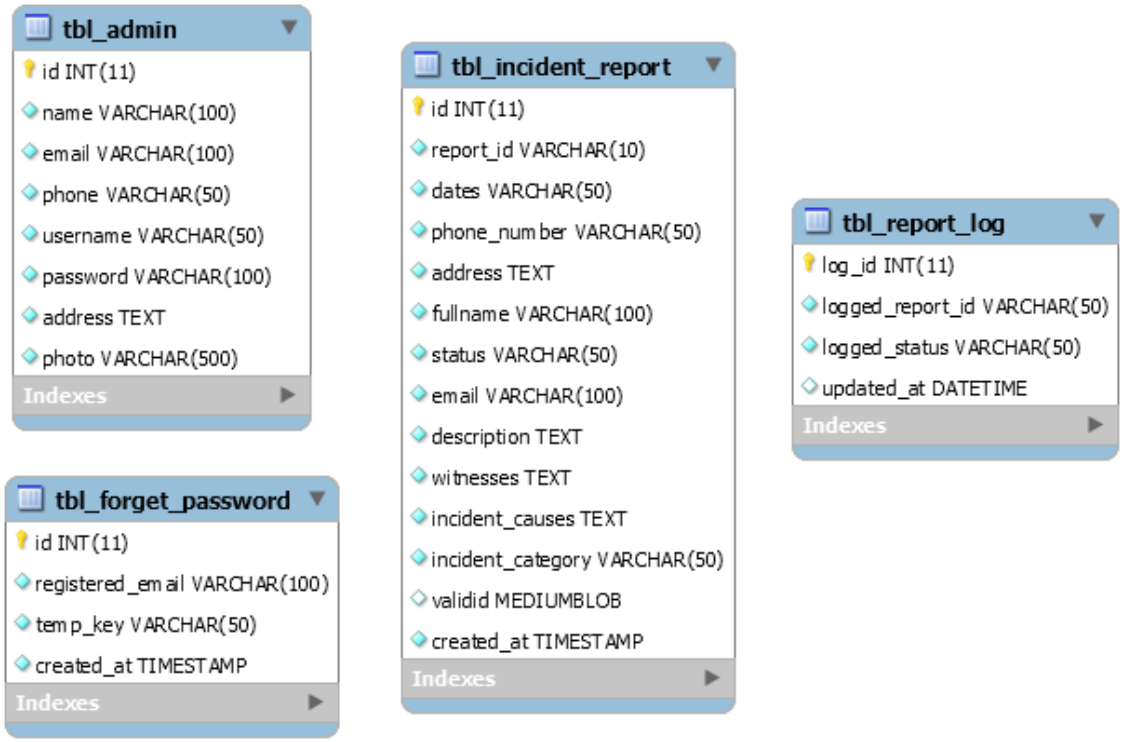


Figure 17. Database Schema

2.2 System Actors

2.2.1 User Roles and Responsibilities / Authority Requirements

User/Role	Example	Frequency of Use	Security/Access, Features Used	Additional Notes
Residents	Phase 1 Residents	Frequent	Residents can access the online platform to report an incident by providing a detailed description of the issue.	None
Committee	Homeowner's Association Grievance and Adjudication Committee	Frequent	Grievance and Adjudication Committee are granted access to the IRMS using the exclusive account. It allows them to view, manage, and track the records to assess the incidents reported by residents.	None

2.3 Dependencies and Change Impacts

2.3.1 System Dependencies

- **File Incident Module –** The Report Management Module relies on the reports of incidents submitted using the File Incident Module. Without user-submitted incidents, the module has no data to manage.

- **Status Transition –** This is dependent on the Pending, Active, Declined, and Archived pages, since it allows incidents to transition between various stages. When an incident is on the Pending page, it can be moved to Active or Declined. When an incident on the Active page is concluded, it can be transferred to the Archived page. These pages are linked to ensure a uniform status flow throughout the system.

- **Admin Access** - Administrators must log in with a master account to access the Report Management Module. Only people with the master account will be able to manage incidents. This guarantees that only authorized users have access to the system.

- **Role Access Control –** The Dashboard Module is dependent on the user role. Only authorized users, such as members of the Grievance and Adjudication Committee, can access the dashboard after logging in. Access to this module is restricted to ensure security and proper data governance.

- **Status Overview** – The Dashboard relies on data from incident report database to display information such as pending, active, and archived incidents. Without a database, the dashboard cannot store or fetch data.

- **Dynamic Updates** – The Dashboard is dependent on real-time synchronization with the Report Management Module to ensure that any status changes in the incident (e.g., from Pending to On-going Incident) are immediately reflected.

- **Report Generation** – The “Generate Report” button depends on Report Management Module that summarizes data into PDF format for record-keeping use.

- **Guest Access –** The File Incident Module is publicly available to residents and administrators who want to file an incident report. Users do not require an account to submit a report, ensure convenience and quick filing of incidents.

- **Data Input –** The functionality of the File Incident Module depends on accurate and complete data provided by the user. It relies on inputs such as the complainant information, including the uploading of valid ID, and incident details.

- **Validation -** Mechanisms are also dependent on this since it ensures that users submit reports with the required fields are not left blank and uploaded valid ID meets the system's format requirements.

- **Submission to Database** – The module relies on the system database to store submitted incident reports. Once a report is submitted, it is saved in the database for review and management by the grievance and adjudication committee via the Report Management Module.

- **Report Management Module** – The File Incident Module directly feeds data to the Report Management Module. Without it, the grievance committee would not have incident reports to review or manage.

2.3.2 Change Impacts

The Homeowners Association and homeowners of Phase 1 San Lorenzo South use the usual method of addressing incident complaints from their community. Previously, residents had to go to the real office to report an incident. Using the proposed system will improve the community's issue reporting and management processes. This approach intends to digitalize the existing method, making it faster, smoother, and more accessible to reports digitally. Overall, this system will modernize and make the process of reporting incidents more accessible and sustainable.

Below is the existing traditional system affected by the proposed system:

1. Filing of Incident Reports
2. Documentation of Incidents
3. Verification of Submitted Incident Reports
4. Communication Between Residents and HOA's Grievance and Adjudication Committee
5. Documentation of Incidents
6. Archiving Reports

3. Functional Specifications

File Incident Report

FUNCTIONALITIES	DESCRIPTION
Incident Form	An online form where residents can input details about the incident, including their personal information, incident category, date, location, and description. Also, uploading an image of their valid ID for verification.
Validation	Ensures all required fields (e.g., full name, phone number, email, valid ID, incident details) are filled out before submission. Invalid or incomplete forms cannot be submitted.
Submission	Send the incident report directly to the system's database for review by user admin or committee members.

Dashboard

FUNCTIONALITIES	DESCRIPTION
Summary Container	Displays quick statistics about the number of pending, on-going, and total incidents in a month. This helps user admin quickly assess the status of reports.
Generate Report	Allows administrators to export a detailed report of incidents in PDF format for documentation or sharing purposes.
Navigation Menu	Provides links to key system modules, such as Pending, Active, Declined, and Archived reports, allowing quick transitions between sections.

Incident Report Management

FUNCTIONALITIES	DESCRIPTION
Incident Report Management Tables	Provides an overview of incident reports, including sorting options for improved organization and administration.
View Details	Provides specific information regarding events, such as contacts, addresses, personal information, persons involved, incident reasons, and incident descriptions.
Pending	These are the incidents that are being reviewed and verified.
Active	Incident reports that are verified and in progress and being actively investigated are located on this page.
Declined	Reports that were declined due to pranks or inadequate information are stored in this table.
Archived	Provides storage to archive the incidents, moving data to the Archives area for recorded history and indicating that the reports have been solved.

Manage Admin Account

FUNCTIONALITIES	DESCRIPTION
Pending Accounts	The Committee Chairman, super admin, can view the list of pending accounts created by the committee official. By accessing this, the super admin can approve or decline the account application according to qualification set by the chairman committee.
Approved Accounts	The super admin can see the activity log of account approvals and the currently active accounts in the system.

Create Admin Account & Reset Password

FUNCTIONALITIES	DESCRIPTION
Create Account	The Committee or admin can create an account by providing necessary details.
Reset Password	Enables the admin to reset their password by verifying their identity through an OTP.

3.1 Nexus: Incident Report Management System

3.1.1 Purpose/ Description

The File Incident Report is a user-friendly online tool that allows residents to report incidents quickly and easily. Instead of visiting the office to file a report, residents can use this module to fill out an incident form, provide their details, describe the incident, and even upload identity verification file like valid ID. This digital method makes it easier and more accessible for residents to share concerns with the Homeowner's Association of Phase 1 San Lorenzo South Subdivision.

This module ensures all submitted reports are organized and sent to the administrators for review. It is a simple yet effective way to collect information, making the process faster and more efficient. By allowing online submissions, it helps reduce paperwork, save time, and ensure that filing incident reports is accessible to everyone in the community.

On the administrative side, The Dashboard gives the user admin an easy way to see the overall status of incident reports in one place. It shows key information like how many reports are on-going, pending, or archived, in a summary container. A graph also helps the administrator see changes over time, like how many incidents happened in the current period compared to before.

It is designed to make managing reports simpler. The user admin can quickly access reports, generate detailed summaries, and analyze data. This module ensures that the administrator stays organized, informed, and can handle incidents more effectively. It is the

central hub for monitoring incident reports and managing user admin, improving how incidents are managed overall.

In addition, Report Management is a tool that provides administrators with a simple yet effective digital platform for managing incident reports. This module modernizes the traditional way of recording incident reports that enhances the transparency, accountability, and efficiency of the operation. There are four categories for incident reports, which include pending, active, declined, and archived, with a status role per incident. The Pending area will ensure that each report submitted is carefully verified before being registered into the active section. This process is very significant in avoiding spam, prank reports, and other fraudulent submitted reports that may affect the operations. Once validated and approved, a report moves to the Active area, where administrators may monitor and handle events closely until they are taken care of. The module, in addition to this, also keeps all the declined reports within the Declined section for the sake of record and reference purposes, even though they were declined throughout the process of verification.

This ensures no submissions are entirely lost, and administrators can go through such reports as and when required. Finally, once the event is resolved fully, it saves it in the Archived area. This archive will be a very good source of historical records, supporting documentation, audits, and future study. The module architecture enhances the administration of the report while ensuring an organized and transparent system that improves overall incident handling operations.

3.1.2 Use case

UC-1	Incident Form Table
Primary Actor(s)	Residents, HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	The residents or committees access the Nexus website interface
Pre-conditions	The residents or committees are connected to the internet to access the website or web application.
Post-conditions	Submits an incident report and files into the user admin system.
Main Success Scenario	1. The resident views the incident from the interface. 2. The resident fills out the required fields of an incident report. 3. The resident submits an incident report.
Extensions	1. The Committee views the incident from interface. 2. The Committee fills out the required fields of an incident report. 3. The Committee submits an incident report.
Priority	High
Special Requirements	The user can be either from and/or filing an incident happened within the Phase 1 San Lorenzo South Subdivision
Open Questions	None

UC-2	User Admin Sign-up
Primary Actor(s)	HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	The committee clicks on a Nexus Admin Page
Pre-conditions	The committee must be connected to the internet to access the website or web application.
Post-conditions	The committee successfully created an account for approval.
Main Success Scenario	1. The committee selects an option to sign up. 2. The committee input required fields such as username, email, and password. 3. The committee will be directed at the email verification page. 4. The committee creates an account and is subject for approval.
Extensions	None
Priority	High
Special Requirements	The user must be strictly authorized committee only to ensure data privacy and security for sensitive information.
Open Questions	None

UC-2	Dashboard Table
Primary Actor(s)	HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	HOA's Administrator accesses the report management interface
Pre-conditions	The administrators are logged into the system to access the interface
Post-conditions	Access the dashboard with summary reports, report generation, and navigation menu.
Main Success Scenario	1. The Committee logs in from the Admin Log-in button 2. The Committee views the Dashboard interface. 3. The Committee views the main menu, graph summary of the reports, generate reports, and the report's statuses.
Extensions	None
Priority	High
Special Requirements	The user must be from the Grievance and Adjudication Committee of the Homeowner's Association of Phase 1 San Lorenzo South Subdivision.
Open Questions	None

UC-3	Incident Report Management Tables
Primary Actor(s)	HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	HOA's Administrator accesses the report management interface.
Pre-conditions	The administrators are logged into the system to access the interface.
Post-conditions	Updates to the tables and reports status are saved in the system.
Main Success Scenario	1. The Committee selects the report management section. 2. The Committee views the interface report management. 3. The Committee updates the status of incident reports.
Extensions	None
Priority	High
Special Requirements	The user must be from the Grievance and Adjudication Committee of the Homeowner's Association of Phase 1 San Lorenzo South Subdivision to ensure data privacy and security for sensitive information.
Open Questions	None

UC-4	View Details
Primary Actor(s)	HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	The administrator clicks on an incident to view the details.
Pre-conditions	The administrator must be logged into the system.
Post-conditions	The administrator successfully views the details of the selected incident.
Main Success Scenario	1. HOA's Administrator logs into the system. 2. HOA's Administrator navigates the incident list. 3. The administrator selects an incident report. 4. The system displays all relevant details of the incident.
Extensions	None
Priority	High
Special Requirements	The user must be from the Grievance and Adjudication Committee of the Homeowner's Association of Phase 1 San Lorenzo South Subdivision to ensure data privacy and security for sensitive information.
Open Questions	None

UC-5	Mediation Hearing Schedule
Primary Actor(s)	HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	The administrator clicks on an incident to view the details.
Pre-conditions	The administrator must be logged into the system.
Post-conditions	The administrator successfully scheduled a mediation hearing for the resident.

Main Success Scenario	1. HOA's Administrator logs into the system. 2. HOA's Administrator navigates the incident list. 3. The administrator selects an incident report. 4. The system displays all relevant details of the incident. 5. The administrator sets a date and time for mediation hearing 6. The system displays a confirmation message and will send a mediation hearing schedule to the residents via email.
Extensions	None
Priority	High
Special Requirements	The user must be from the Grievance and Adjudication Committee of the Homeowner's Association of Phase 1 San Lorenzo South Subdivision to ensure data privacy and security for sensitive information.
Open Questions	None

UC-6	**Report Management Module**
Primary Actor(s)	HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	The administrator navigates to the Reports Management section and selects a specific incident report type (Pending, Active, Declined, Archive)
Pre-conditions	Administrators must be logged into the system.
Post-conditions	Administrators can view incidents based on the selected status (Pending, Active, Declined, Archive)
Main Success Scenario	1. HOA's Administrator logs into the system. 2. HOA's Administrator navigates to the side menu bar section which includes "Pending", "Active", "Declined", and "Archive". 3. The administrator selects one of the following options: a. **Pending**: View new incidents that are under review for verification. b. **Active**: View incidents that have been accepted and are currently being managed. c. **Declined**: View incidents that were not accepted, stored for future reference. d. **Archive**: View completed or archived incidents for historical context. 4. The system displays a list of incidents for the selected status. 5. Administrator clicks on an incident to view its details. 6. HOA's Administrator returns to the list of incidents. 7. The Administrator clicks the "logout" button to exit the system.
Extensions	If there are no incidents in the selected category, display a message indicating this.
Priority	High
Special Requirements	The system should allow filtering and sorting of pending incidents.

Open Questions	None

UC-7	Settlement Section
Primary Actor(s)	HOA's Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Grievance and Adjudication Committee
Trigger	The administrator clicks on an incident to view the details.
Pre-conditions	The administrator must be logged into the system.
Post-conditions	The administrator successfully solved the incident report.
Main Success Scenario	1. HOA's Administrator logs into the system. 2. HOA's Administrator navigates the incident list. 3. The administrator selects an incident report. 4. The system displays all relevant details of the incident. 5. The administrator updates the incident report with the settlement summary for finalization of the report. 6. The system displays confirmation message and will archive the active incident report.
Extensions	None
Priority	High
Special Requirements	The user must be from the Grievance and Adjudication Committee of the Homeowner's Association of Phase 1 San Lorenzo South Subdivision to ensure data privacy and security for sensitive information.
Open Questions	None

UC-8	Manage Admin Account
Primary Actor(s)	HOA's Chairman of Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Chairman of Grievance and Adjudication Committee
Trigger	The super admin clicks on the manage admin menu
Pre-conditions	The chairman must be logged into the system as super admin.
Post-conditions	The super admin successfully updates the account status.
Main Success Scenario	1. Super admin navigates the pending accounts. 2. The super admin views the account settings. 3. The super admin will approve the account application and set the account as admin.
Extensions	1. Super admin will decline the account application and will send the reason for account rejection to the account owner.
Priority	High
Special Requirements	The user must be from the Grievance and Adjudication Committee of the Homeowner's Association of Phase 1 San Lorenzo South Subdivision to ensure data privacy and security for sensitive information.

Open Questions	None
UC-9	**Create Account and Reset Password**
Primary Actor(s)	HOA's Chairman of Grievance and Adjudication Committee
Stakeholders and Interest	System User Administrator, HOA's Chairman of Grievance and Adjudication Committee
Trigger	The super admin clicks on the manage admin menu
Pre-conditions	The chairman must be logged into the system as super admin.
Post-conditions	The super admin successfully updates the account status.
Main Success Scenario	1. Super admin navigates the pending accounts. 2. The super admin views the account settings. 3. The super admin will approve the account application and set the account as admin.
Extensions	1. Super admin will decline the account application and will send the reason for account rejection to the account owner.
Priority	High
Special Requirements	The user must be from the Grievance and Adjudication Committee of the Homeowner's Association of Phase 1 San Lorenzo South Subdivision to ensure data privacy and security for sensitive information.
Open Questions	None

3.1.3 Mock-up



Figure 18. Guest Resident Module – Landing Page

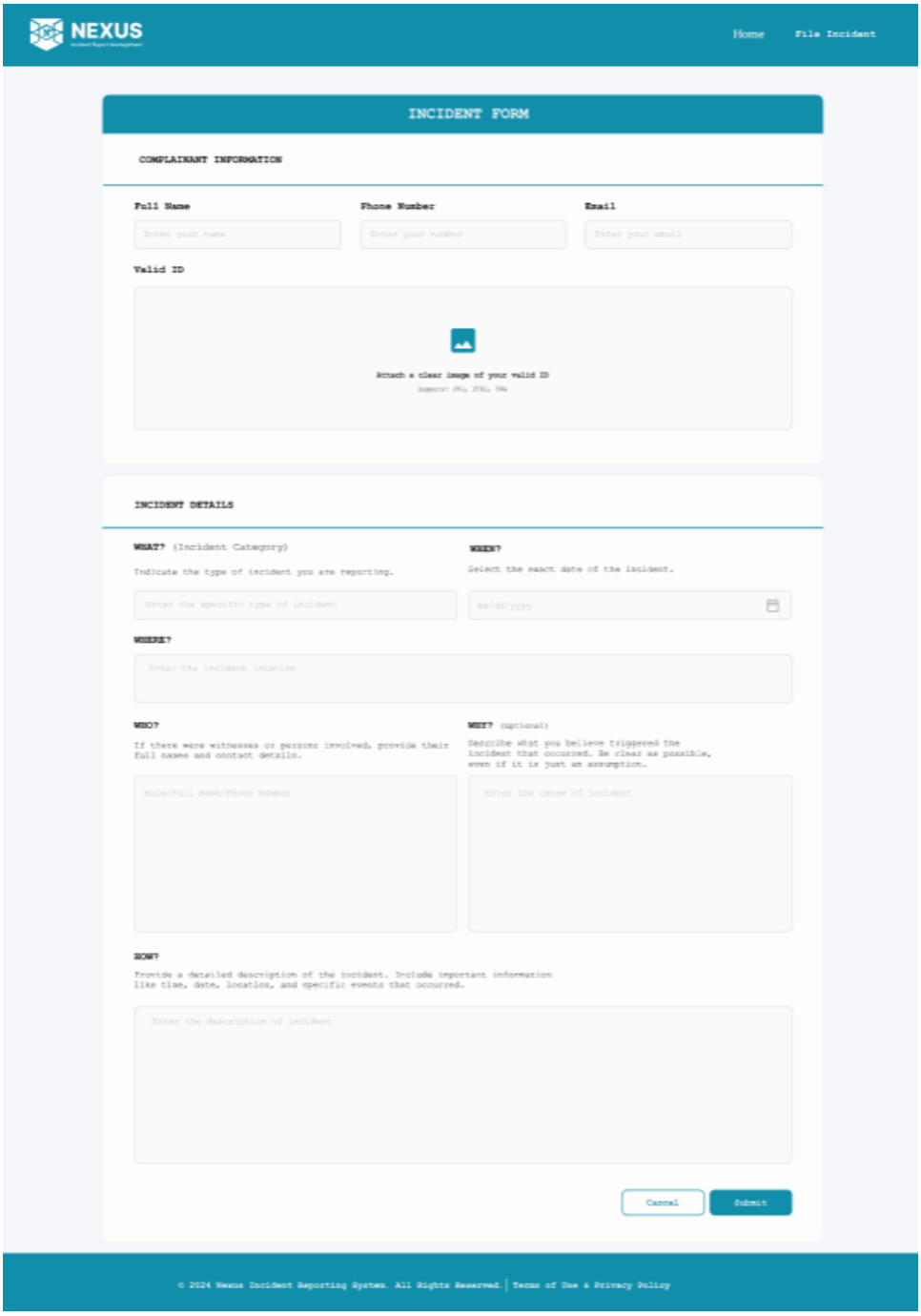


Figure 18. Guest Resident Module – File Incident Page

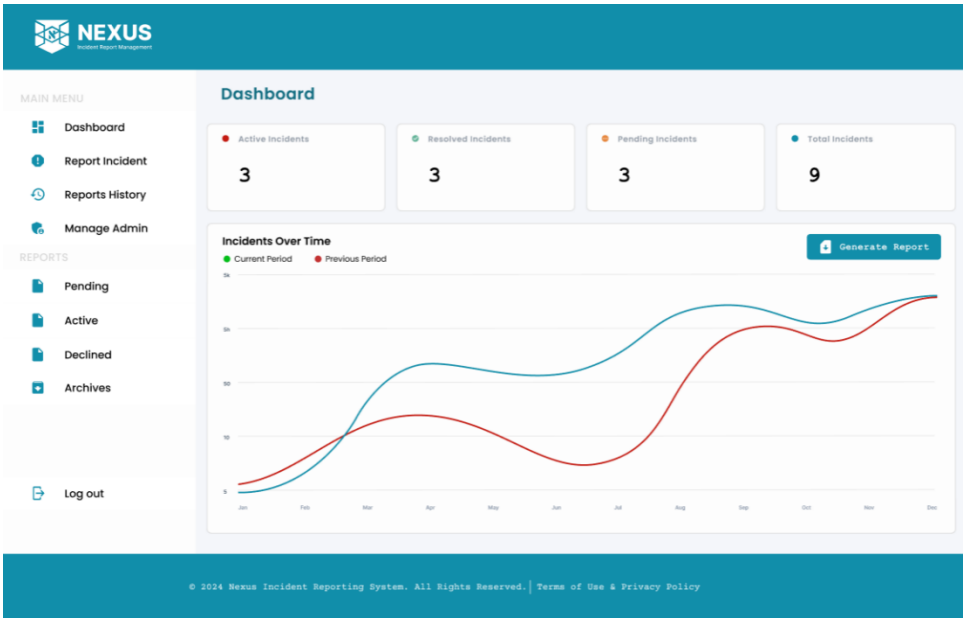


Figure 19.7 Super admin – Dashboard Page

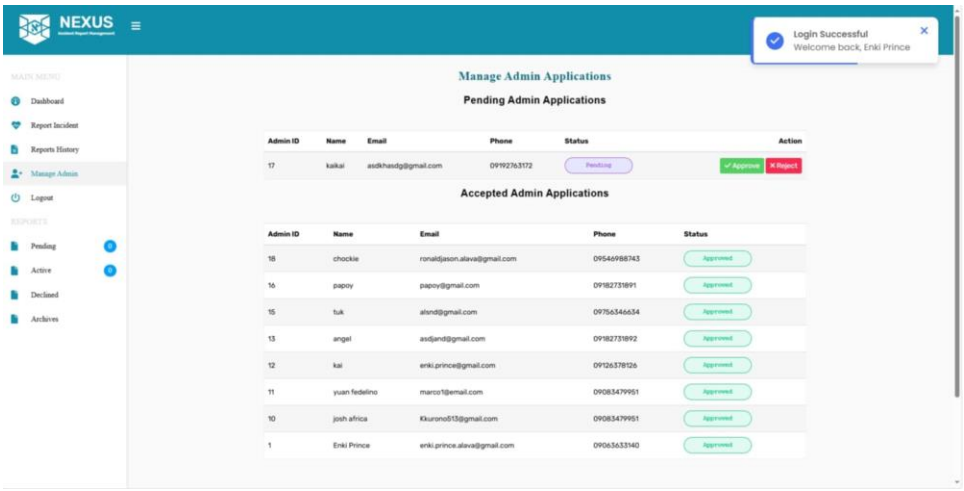


Figure 19.7 Super admin – Manage Admin Page

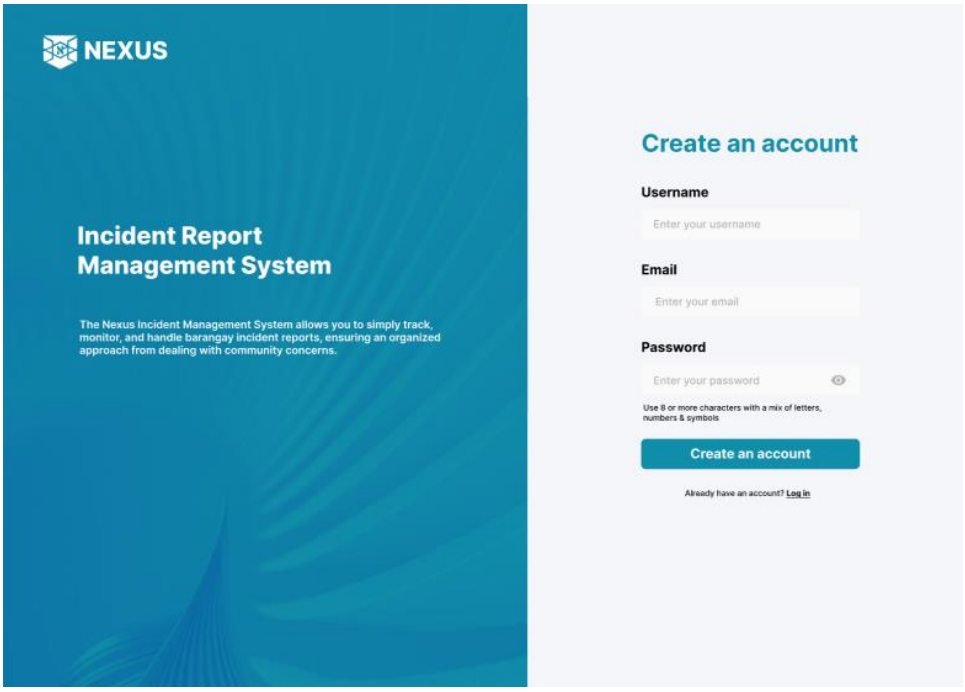


Figure 18. User Admin Registration Module – Create an Account Page

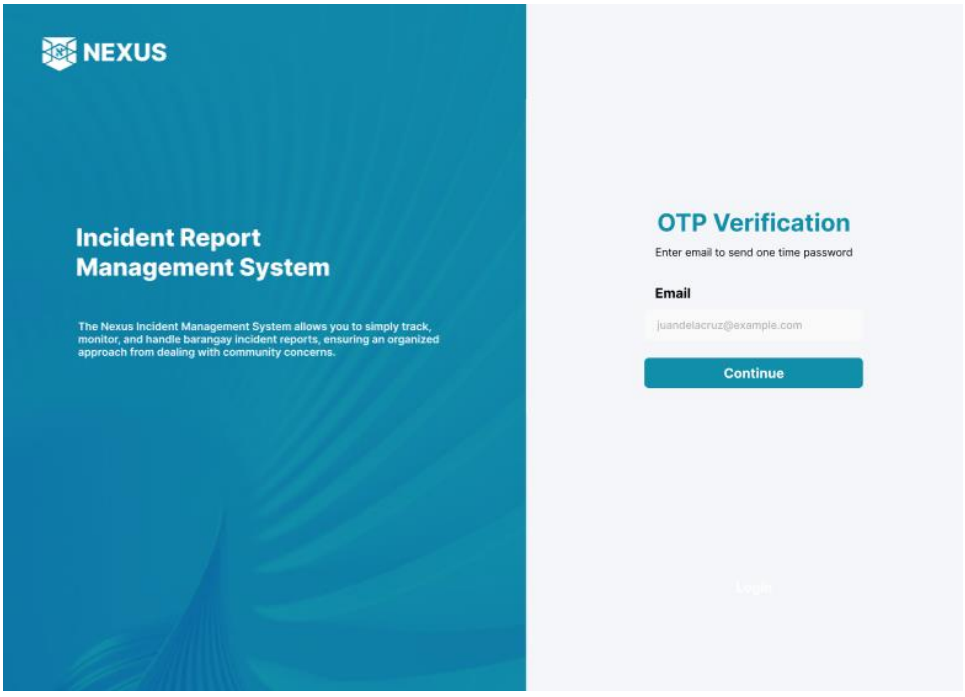


Figure 18.1 User Admin Registration Module – Email Verification via OTP

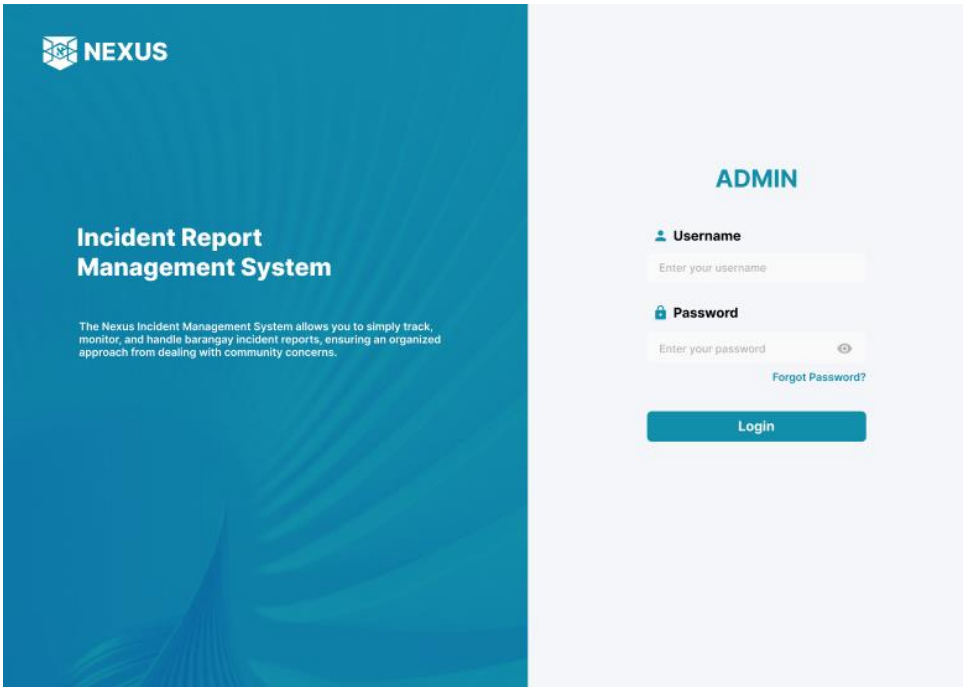


Figure 19. User Admin – Sign-in Page

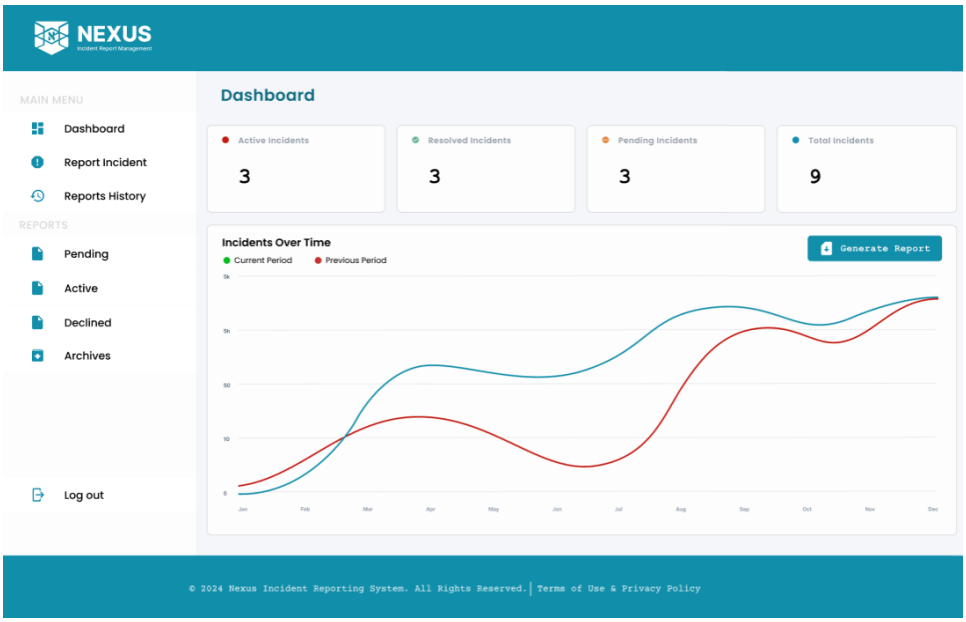


Figure 19.1 User Admin – Dashboard Page

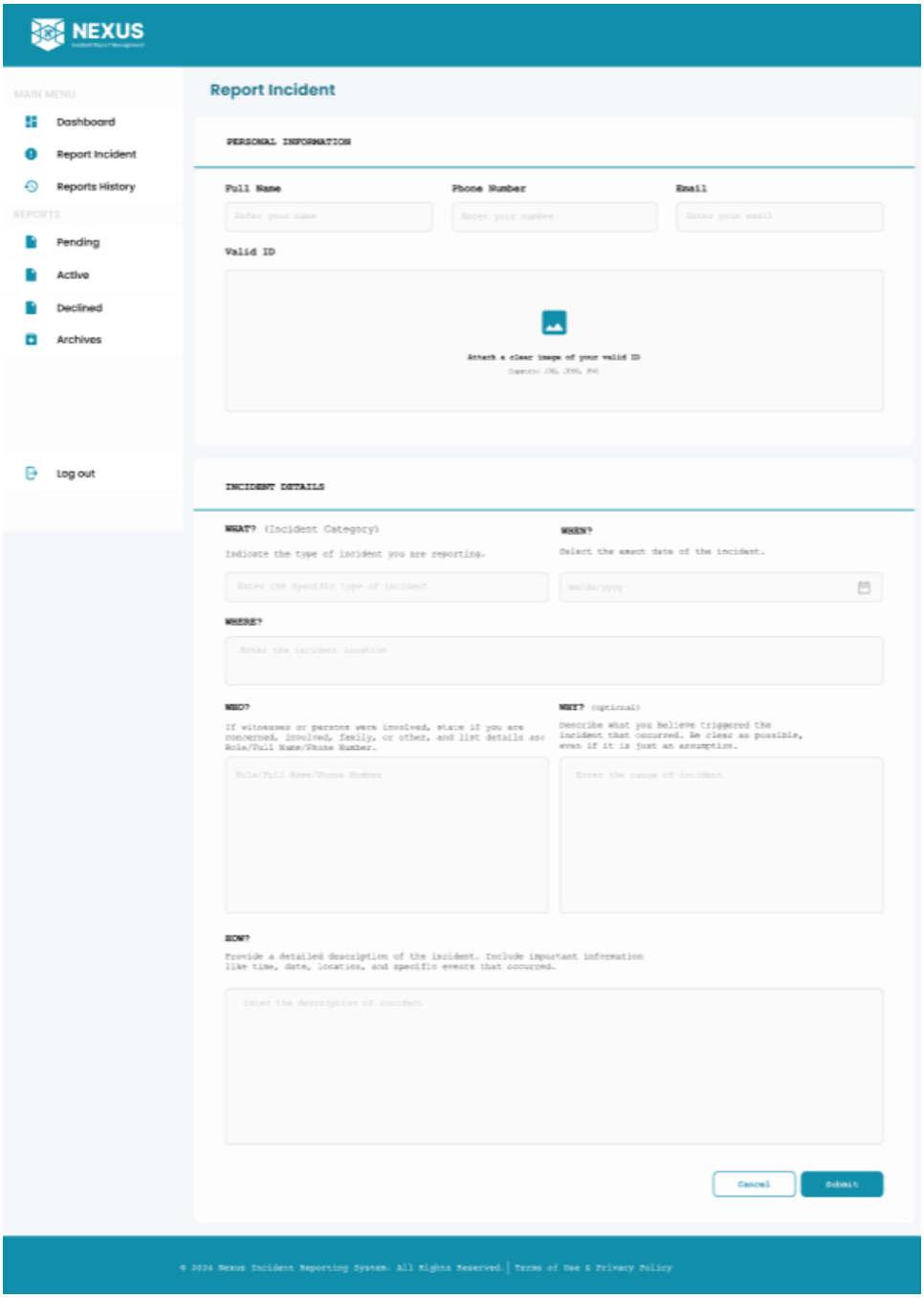


Figure 19.2 User Admin – Report Incident Walk-in Page

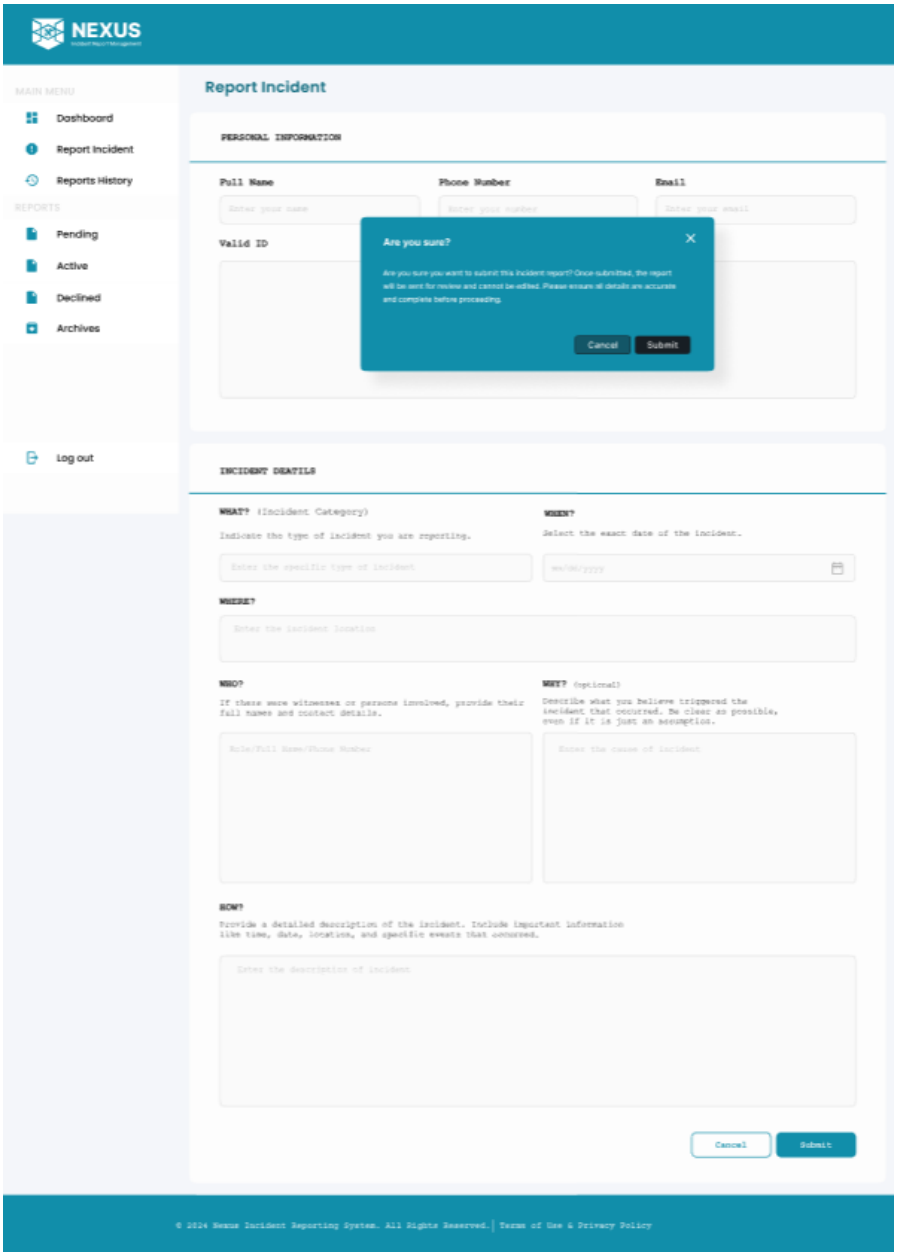


Figure 19.2 User Admin – Report Incident Walk-in Submit Confirmation

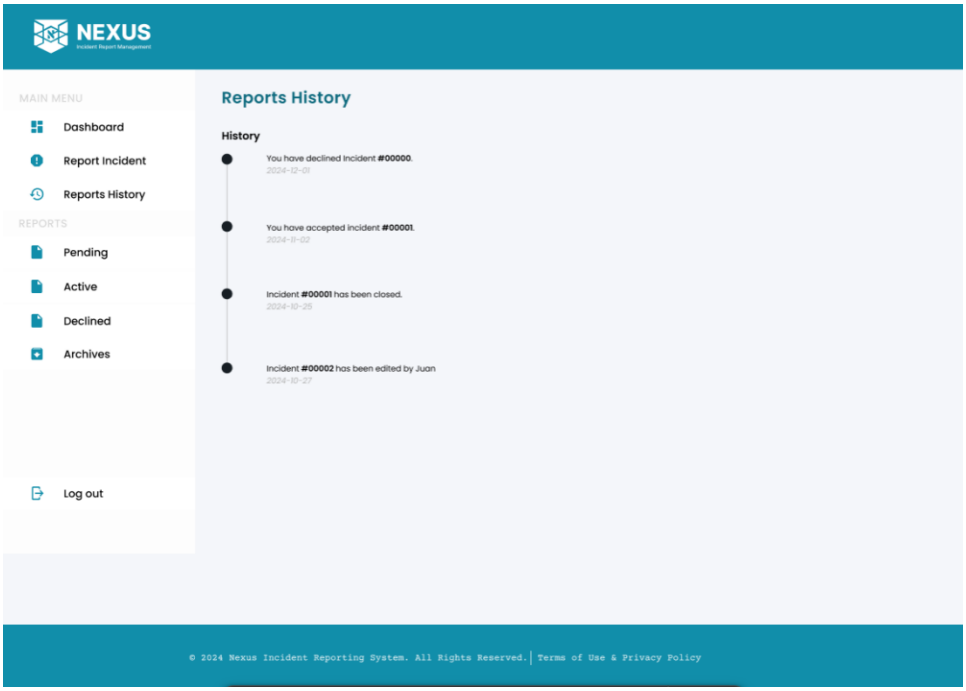


Figure 19.3 User Admin – Report History Page

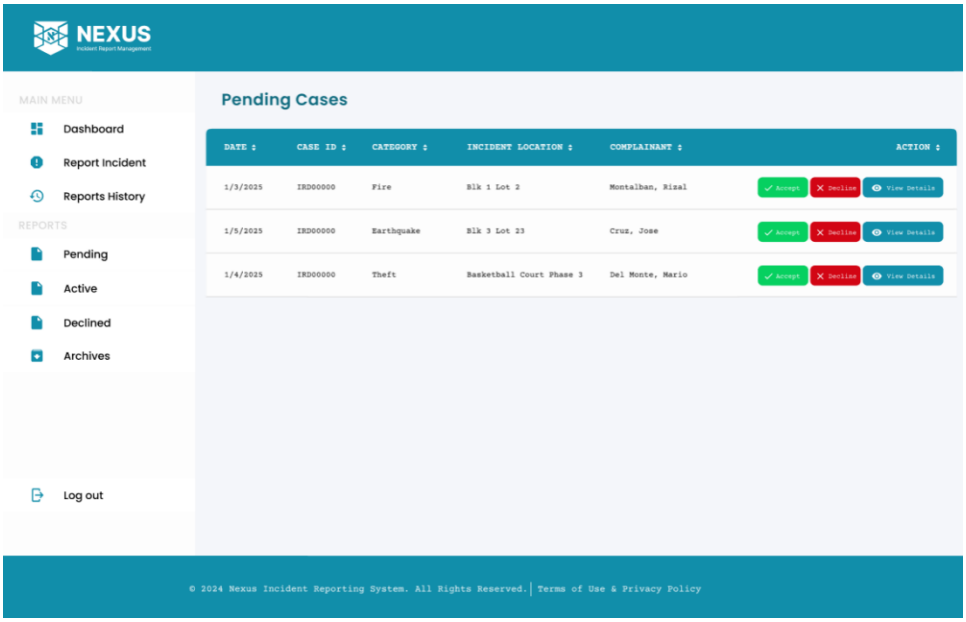


Figure 19.4 User Admin – Pending Incident Reports

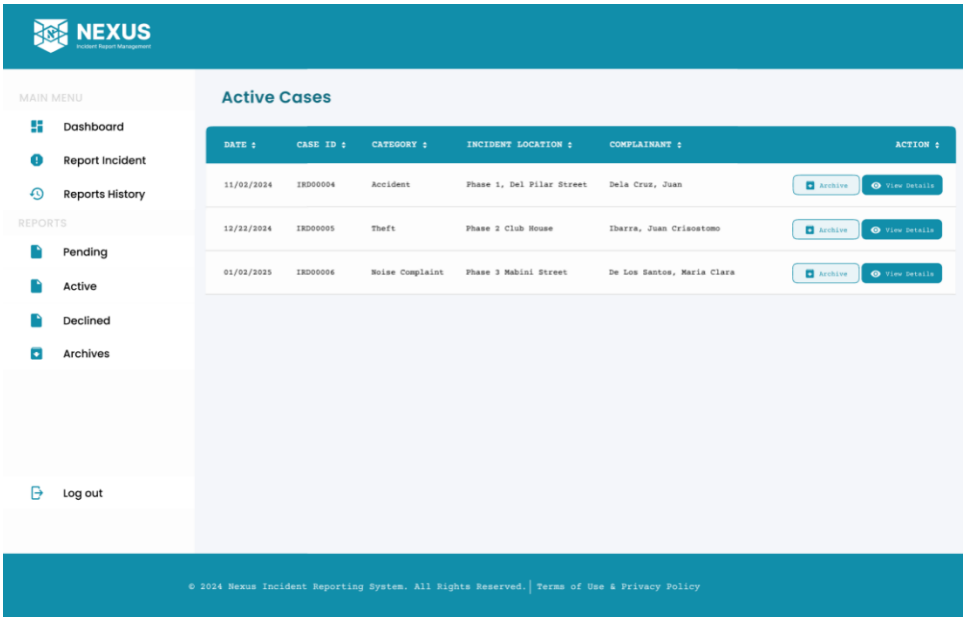


Figure 19.5 User Admin – Active Incident Reports

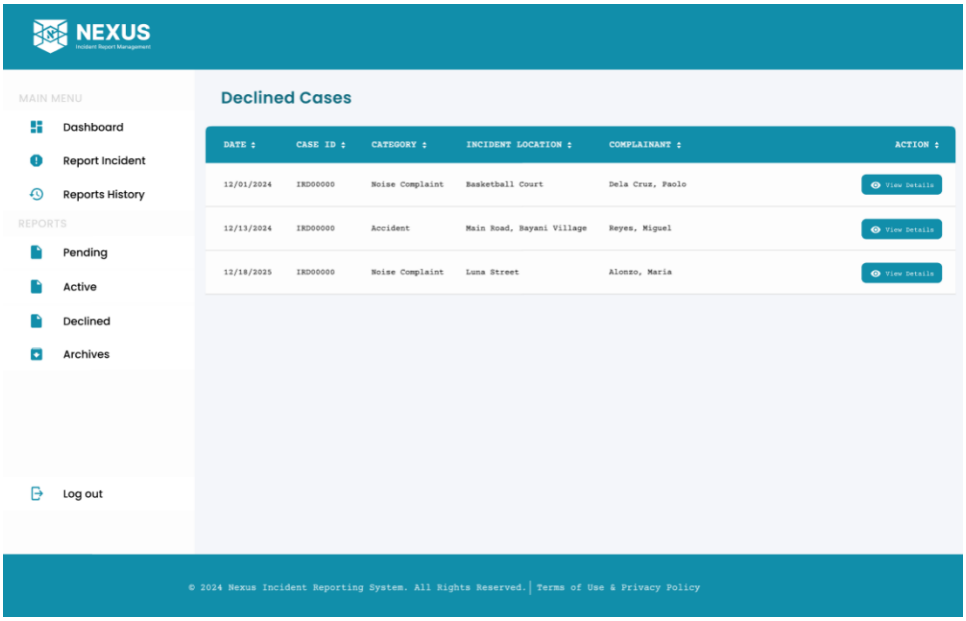


Figure 19.6 User Admin – Declined Incident Reports

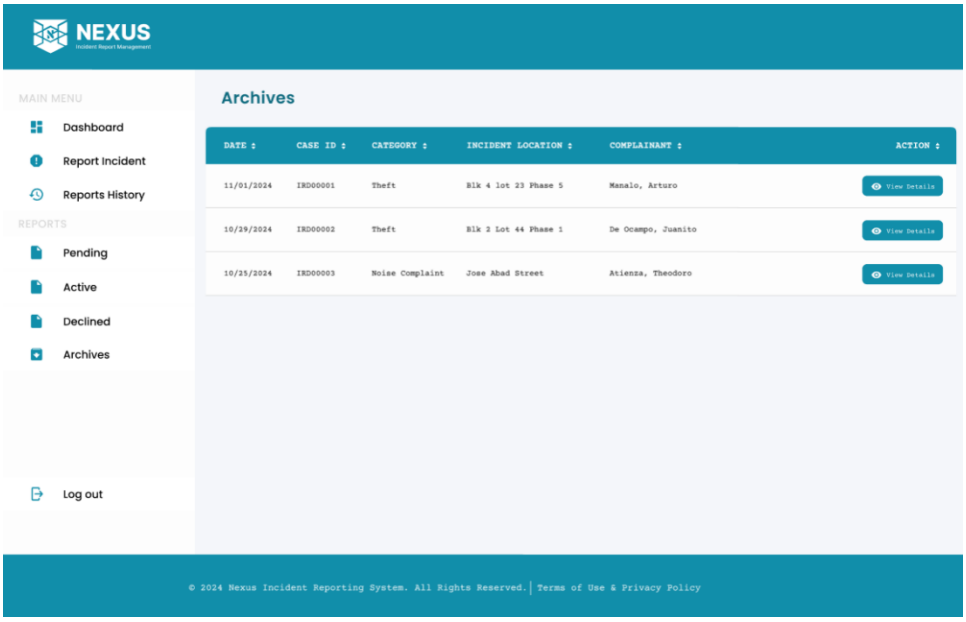


Figure 19.7 User Admin – Archive Incident Reports

3.1.4 Functional Requirements

Spec ID	Specification Description	Business Rules/ Data Dependency
1	Username for user admin registration	Username must be the subdivision of the homeowner's association
2	Email address for user admin registration	Email format validation must be unique
3	Password for registration	Must be at least 8 characters and strong
4	Confirm Password for user admin account	Must match the password field
5	Submits the registration form	All fields must be valid before enabling submission
6	Full Name of the guest	Must be characters
7	Phone number of the guest	Must be 11-digit number
8	Email address of the guest	Email format validation must be unique
9	Upload valid identification	File must be an image
10	Incident Category	Cannot be empty
11	Exact date of the incident	Date format validation must be a previous and present date only and valid year.
12	IRMS allows users to submit incident report online	All fields must have inputs and are accurate
13	Review and verify the submitted incident	Admins verify the reports if it is accepted or will be declined
14	The reports can be stored to archive after resolution	Archived reports are accessible for future references

3.1.5 Field level specifications

Form Elements:

Sign Up/Login/Reset Password Elements

Call-out	Field Label	UI Control	Mand?	Editable ?	Data Type	Value Set	Default Value	Data Example	Data Source
	Username	Input Field	Yes	Yes	String	-	-	Juan Cruz	User input
	Email	Input Field	Yes	Yes	String	-	-	Juancruz @example.com	User input
	Password	Input Field	Yes	Yes	String	-	-	*******	User input
	New Password	Input Field	Yes	Yes	String	-	-	*******	User Input
	Confirm Password	Input Field	Yes	Yes	String	-	-	*******	User Input
	OTP Verification	Input Field	Yes	Yes	String	-	-	012398	User Input
	Create Account	Button	Yes	Yes	No	-	-	-	-
	Continue	Button	Yes	Yes	No	-	-	-	-

File Incident Report Elements

Call-out	Field Label	UI Control	Mand?	Editable	Data Type	Value Set	Default Value	Data Example	Data Source
	Full Name	Input Field	Yes	Yes	String	–	-	Juan Dela Cruz	User input
	Phone Number	Input Field	Yes	Yes	String	–	-	+63XXXXXXXXX	User input
	Email	Input Field	Yes	Yes	String	–	-	example@domain.com	User input
	Valid ID	File Upload	Yes	Yes	File	–	JPEG & PNG	Image.jpeg	User input
	What?	Input Field	Yes	Yes	String	–	-	Noise Complaint	User input

	When?	Input Field	Yes	Yes	Date and Time	–	-	01/02/2025 4:37 AM	User input
	Where?	Input Field	Yes	Yes	String	–	-	Narra St, Phase 1	User input
	Who?	Input Field	Yes	Yes	String	–	-	Maria Ibarra	User input
	Why?	Input Field	No	Yes	String	–	-	Noise	User input
	How?	Input Field	Yes	Yes	String	–	-	Loud music from neighboring apartment all night	User input
	Submit	Button	Yes	No	–	–	-	–	–

Form Business Rules and Dependencies:

Field Label	Validation / Business Rules	Error Messages	Data Dependencies	Additional Info/ Notes
Full Name		Please fill out this field	None	
Phone Number	Should be valid phone number.	Invalid please enter a valid number	None	
Email	Should be a valid email	Please fill out this field	None	
Valid ID	The file must be JPG or PNG	Invalid file type. Only JPG and PNG are allowed.	None	
What?		Please fill out this field	None	
When?		Please fill out this field	None	
Where?		Please fill out this field	None	
Who?		Please fill out this field	None	
Why?		Please fill out this field	None	
How?		Please fill out this field	None	

Buttons, Links and Icons:

Button, Link, Icon Label	OnClick Event	Other Event	Visible	Enabled Vs Disabled	Navigate To	Validation	Dependencies
Home	Lands on a home page	None	Yes, always	Enabled, by default.	Home Page		
File incident	Redirects to Incident Report Form	None	Yes, always		File Incident		
Submit	Verify if all required fields have inputs. If yes, submit the form.	After clicking submit, the confirmation message prompts “Are you sure you want to submit? Ensure all details are accurate and complete before proceeding”	Yes, always	Enabled, by default.	N/A	Required all fields must be accomplished to submit	File Incident: Full name, Phone number, Email, Valid ID, What, When, Where, and How
Generate Report	Triggers the report generation process	None	Yes, always	Enabled, by default	N/A		
Accept	Marks a report as accepted and moves it to Active section	Triggers the report accepted notification	Visible only in Pending section	Enabled only for reports on verification process	N/A	Ensures the reports is valid and is not some sort of attack	Requires a pending report
Decline	Marks a report as declined and moves it to Declined section	Notifies the residents and triggers the report declined notification	Only visible in Pending section	Enabled only for reports on verification process	N/A	Reason for decline is provided	Requires a pending report
View Details	Opens the detailed view of selected report	Expand the report details in a separate page	Always visible for all reports	Always enabled	Details report view page	No validation required	Requires a selected report

Archive	Move a report to the archives	Only visible in the Active section	Enabled only for reports that are solved	Enabled only for solved reports	Remains on the same page	Ensures the report is solved before archiving	Requires an active report
Dashboard	Navigates to dashboard	N/A	Always visible	Always enabled	Dashboard page	No validation required	N/A
Reports History	Opens the report's history page	N/A	Always visible	Always enabled	Reports history page	No validation required	N/A
Pending	Shows the report	N/A	Always visible	Always enabled	Reports pending page	No validation required	N/A
Active	Navigates to active section	N/A	Always visible	Always enabled	Reports active page	No validation required	N/A
Declined	Show only the declined reports	N/A	Always visible	Always enabled	Reports declined page	No validation required	N/A
Archives	Shows the archived reports	N/A	Always visible	Always enabled	Reports archives page	No validation required	N/A

4. System Configurations

1. Installation

- **Purpose:** Install browsing software and prepare components needed for setup to function.
- **Alternatives:** Use different deployment methods and platforms
- **Customization**: Modifying the interface of the application to align with association's branding.
- **Dependencies:** Ensure compatibility with the web servers, data servers, and network firewalls.

2. Admin Account Management

- **Purpose:** Define the roles within the IRMS to ensure proper access control and functionality.
- **Alternatives:** Use a custom authentication mechanism
- **Customization**: Modify the permission associated with each role based on association's needs.
- **Dependencies:** Backend logic that enforces permission based on user roles.

3. **Incident Reporting**

- **Purpose:** Ensures that all necessary information is collected for efficient tracking and resolution.
- **Alternatives:** Using a simple report form.
- **Customization**: Custom fields for incident reports and attachments options.
- **Dependencies:** Frontend form validation, backend process logic, database schema for incident reports.

4. **Database Configuration**

- **Purpose:** To set up database backend for storing the system's data of incident reports.
- **Alternatives:** Choose a good and secure database like MySQL and SQL server.
- **Customization**: Configure the database parameters and tables.
- **Dependencies:** Ensure the proper connectivity of database for accuracy.

5. Other System Requirements/ Non-Functional Requirements

- **Security** - The system must implement strict security measures to prevent unauthorized access and data breaches. Only the authorized Grievance and Adjudication committee should be able to manage incident reports, ensuring the confidentiality of sensitive information.
- **Integrity -** The system should keep incident reports safe and unchanged unless updated by authorized committees. All actions should be recorded for tracking.
- **Availability** - The system must be accessible to users at all times to allow residents to file incident reports conveniently and enable committees to manage reports efficiently.
- **Identity Verification** - Residents must verify their identity by uploading a valid ID upon submission of an incident report. Committees must log in using secure credentials to access and manage reports.
- **Performance** - The system should operate efficiently, allowing quick submission, retrieval, and processing of incident reports. It should be optimized to handle multiple users and ensure minimal downtime.
- **Usability** - The user interface should be intuitive, easy to navigate, and accessible to users with different levels of technical expertise. Residents should be able to submit reports with minimal steps, and committees should have clear workflows for managing incidents.

6. Reporting Requirements

Report: Incident Report

Scope: This will provide an overview of the incident report details that the residents filled out including personal information and description of the incidents. This also includes incident ID, report status, and can generate PDF file on the part of administrators.

Format: The report will be a statistic, a categorized incident list and graph.

Data Elements and Contents Required: The report includes the following

- Incident ID
- Date
- Incident Information
- Incidents Statuses

7. Integration Requirements

Integration of Email API – Twilio SendGrid

The system integrates Twilio SendGrid to handle email notifications efficiently. This API ensures that residents receive email confirmations after resolving an incident report, keeping them informed about the outcome of the incident by settlement. Committee officials also use email to verify or retrieve their account via OTP. This integration improves communication and ensures transparency between residents and committees.

Integration of Calendar API – Google Calendar

To enhance scheduling and incident tracking, the system integrates Google Calendar API. The integration is initiated with the required field “When?” for the residents to select a date of the incident. This also allows committee officials to set a schedule for the mediation hearing and further investigations. By linking with calendar events, the system ensures that no incident reports are overlooked, improving efficiency in handling community issues.

7.1 Exception Handling/ Error Reporting

Exception /Error ID	Error	Cause	Solution Strategy
ERR001	Password validations not displayed on login	Missing validation messages	Ensure validation messages are triggered and displayed properly
ERR002	Table shows "1 of 1" even with no results	Incorrect logic in data display	Adjust table logic to show "0 of 0" when no records exist
ERR003	Leading spaces not validated in fields	Input validation missing for leading spaces	Implement trim function and input validation
ERR004	Responsiveness issue in all pages except sign-up and login	CSS and media queries not properly implemented	Apply responsive design techniques across all pages
ERR005	Incorrect phone number placeholder (resident/admin incident form, sign-up)	Mismatched format in input placeholder	Update placeholder text to match required phone format
ERR006	BCE past years accepted in date fields (incident forms)	Date validation missing for BCE years	Implement validation to restrict BCE years input
ERR007	Notification counter does not disappear when there are no pending reports	Counter update logic is incorrect	Modify logic to hide the counter when there are no pending reports
ERR008	Sidebar and dashboard width do not minimize when clicking the hamburger button; icons are inconsistent	Missing event handling for sidebar toggle and inconsistent icon usage	Implement event listeners for the hamburger button and standardize sidebar icons
ERR009	After successful password reset, login page displays an error container	Incorrect state management or unnecessary error display	Ensure error messages reset upon successful password reset
ERR010	Successful login allows navigation back to the sign-in page without destroying session	Session not properly managed on login	Implement session handling to restrict access to the sign-in page after login
ERR011	Login attempts validation missing	Security mechanism not implemented	Implement login attempt tracking and lockout mechanisms

8. Data Migration/ Conversion Requirements

In the framework of the Nexus Incident Report Management System (NIRMS) for tracking and managing incidents efficiently within an organization, the data migration/conversion requirements involve migrating and merging critical data from multiple sources into the NIRMS platform to ensure seamless operations and alignment with organizational goals.

Objectives:

This system development project aims to establish a comprehensive incident report management system tailored for organizations. The system will efficiently handle incident reporting, categorization, tracking, resolution management, and generating reports. The product will be a user-friendly, web-based platform accessible to administrators, employees, and relevant stakeholders, ensuring streamlined and transparent incident management processes.

Scope:

Implement and develop a Nexus Incident Report Management System to enhance incident tracking and resolution processes. The system will replace existing procedures with a centralized and integrated platform for managing incident data, including incident logging, categorization, prioritization, status tracking, resolution updates, and reporting. This approach will ensure efficient data flow and enhanced accountability.

Data Mapping:

A data mapping document will define how data from each source (e.g., spreadsheets, legacy systems, or manual logs) will be mapped to the corresponding fields in the NIRMS. This ensures data accuracy and consistency during migration, aligning existing data formats with the system's structure and requirements.

Data Extraction:

Relevant incident data, including incident descriptions, timestamps, locations, reporters, and resolutions, will be extracted from identified sources. This may involve exporting data from databases, spreadsheets, or other legacy systems, ensuring all pertinent information is captured for migration.

8.1 Data Conversion Strategy

Data Extraction

- Identify the Source: Identify where the original incident report data resides, such as legacy systems, spreadsheets, text files, or databases.
- Extract the Data: Use appropriate tools like SQL queries, APIs, or manual entry to extract relevant data, including incident descriptions, timestamps, reporter information, and resolution statuses.

Data Transformation

- Clean Up the Data: Removes inconsistencies, duplicates, and errors by standardizing formats (e.g., dates, text fields) and eliminating irrelevant or redundant information.
- Map the Data: Defines the correspondence between fields in the old system and those in the new NIRMS schema. For example, maps "Reporter Name" to "Username" or "Incident Type Code" to "Incident Category."
- Transform the Data: Convert data types as required, combine or split fields (eg, separating full names into first and last names), and structure the data in formats compatible with NIRMS.

Data Loading

- Prepare the Target System: Set up the NIRMS platform by creating necessary tables, fields, and relationships for importing data. Ensure the system is ready to receive and validate incoming data.
- Load the Data: Import the transformed data into NIRMS using ETL tools or manual import procedures while minimizing disruptions to the system's performance and availability.
- Verify the Load: Conduct queries and validations to ensure that all data has been accurately and completely imported, matching record counts and checking for anomalies.

Conversion Schedule

1. Planning

- Defines the scope of the migration.
- Identify all datasets to be migrated.
- Create a detailed data conversion plan with a timeline and responsibilities.

2. Data Extraction

- Extract data securely from the existing system.
- Store the data in an accessible, secure location for transformation.

3. Data Transformation

- Clean the extracted data to eliminate inconsistencies and errors.
- Map the data fields from the old system to the new system.

- Convert the data to a format accepted by NIRMS.

4. Data Loading

- Import the transformed data into NIRMS.
- Verify that all data is correctly imported and cross-check for accuracy.

5. Testing

- Validate the integrity of the imported data by performing test cases.
- Ensure data is functional in workflows, including incident tracking and reporting.

6. Review and Adjustments

- Assess the overall migration process.
- Address any discrepancies or errors discovered during testing.
- Prepare for the final live conversion, ensuring a seamless transition.

8.2 Data Conversion Preparation

Preparing for data conversion is a critical phase in ensuring the smooth transfer of information, especially in scenarios where challenges may arise during the transition of incident data within the Nexus: Incident Report Management System (NIRMS). This preparation involves proactive planning and implementing strategies to address potential issues and enable an effective data conversion process.

Key Steps in Data Conversion Preparation:

1. Evaluate Requirements

- Assess the specific data conversion needs for managing incident reports.
- Analyze the current data landscape, including data quality, format, structure, and relevance, to facilitate a smooth migration to the new Incident Report Management System (IRMS).

2. Identify Root Causes of Conversion Failures

- Investigate the factors causing unsuccessful data conversions, such as technical limitations, data quality issues, mapping errors, or compatibility challenges between the source and target systems.

3. Assess the Impact of Conversion Failures

- Determine how failed data conversions affect the incident reporting process, including disruptions in data accessibility, inaccuracies in report handling, and challenges for users in utilizing the system effectively.

4. Examine the Problematic Data

- Conduct a detailed analysis of the data involved in conversion failures.
- Validate the data's accuracy, consistency, and completeness, and identify any errors or anomalies that may have contributed to the issues.

5. Establish a Backup Strategy

- Develop a robust backup plan to safeguard all incident-related data, such as reports, logs, and critical information.
- Ensure that backups are encrypted, secure, and readily available for recovery in case of data loss or system failures.

6. Implement a Restoration Procedure

- Create a restoration process to reinstate system functionality while maintaining data accuracy and integrity.
- Confirm that backups are up-to-date, accessible, and securely stored before initiating restoration procedures.

7. Engage Stakeholders

- Communicate with relevant stakeholders, including incident managers, report handlers, and system administrators, about the data conversion process.
- Provide necessary training and support to ensure stakeholders can effectively interact with the converted data.

8. Allocate Resources

- Assign adequate human, financial, and time resources to the data conversion process.
- Ensure the team executing the conversion possesses the required skills and expertise.

9. Verification and Validation

- Create a testing environment to validate the converted data.
- Conduct comprehensive tests to ensure the data meets quality standards and aligns with the operational requirements of the Incident Report Management System.
- Perform sample checks and comparisons with existing records to confirm data accuracy.

By following these steps, organizations can prepare effectively for data conversion within the Incident Report Management System, minimizing risks and ensuring a smooth transition that maintains the integrity and usability of critical incident data.

8.3 Data Conversion Specifications

Source	Source Data Element	Target	Target Data Element	Conversion Rules	Notes
Resident	Phone Number	Incident Report	Phone Number	Direct mapping	No transformation needed
Resident	Full Name	Incident Report	Full Name	Direct mapping	No transformation needed
Resident	Email	Incident Report	Email	Direct mapping	No transformation needed
Resident	Incident Category (What)	Incident Report	Incident Category (What)	Direct mapping	Standardized list of categories must match
Resident	Incident Location (Where)	Incident Report	Incident Location (Where)	Direct mapping	Location data should be verified for accuracy
Resident	Incident Description (How)	Incident Report	Incident Description (How)	Direct mapping	Ensure text formatting is retained
Resident	Incident Causes (How)	Incident Report	Incident Causes (How)	Direct mapping	Ensure multiple causes are separated properly
Resident	People Involved (Who)	Incident Report	People Involved (Who)	Direct mapping	List format should be retained

9. Open Issues

Issue ID	Issue	Raised By	Raised On	Solution/ Decision	Resolved By	Resolved On	Status
ERR001	Password validations not displayed on login	Patrick Jeremie Maniaul	1/25/2025	Ensure validation messages are triggered and displayed properly	Enki Prince	1/30/2025	Passed
ERR002	Table shows "1 of 1" even with no results	Patrick Jeremie Maniaul	1/25/2025	Adjust table logic to show "0 of 0" when no records exist	Enki Prince	1/30/2025	Passed
ERR003	Leading spaces not validated in fields	Patrick Jeremie Maniaul	1/25/2025	Implement trim function and input validation	Enki Prince	1/30/2025	Passed
ERR004	Responsiveness issue in all pages except sign-up and login	Patrick Jeremie Maniaul	1/25/2025	Apply responsive design techniques across all pages	Enki Prince	1/30/2025	Passed
ERR005	Incorrect phone number placeholder (resident/admin incident	Patrick Jeremie Maniaul	1/25/2025	Update placeholder text to match required phone format	Enki Prince	1/30/2025	Passed

	form, sign-up)						
ERR006	BCE past years accepted in date fields (incident forms)	Patrick Jeremie Maniaul	1/25/2025	Implement validation to restrict BCE years input	Enki Prince	1/30/2025	Passed
ERR007	Notification counter does not disappear when there are no pending reports	Patrick Jeremie Maniaul	1/25/2025	Modify logic to hide the counter when there are no pending reports	Enki Prince	1/30/2025	Passed
ERR008	Sidebar and dashboard width does not minimize when clicking the hamburger button; icons are inconsistent	Patrick Jeremie Maniaul	1/25/2025	Implement event listeners for the hamburger button and standardize sidebar icons	Enki Prince	1/30/2025	Passed
ERR009	After successful password reset, login page displays an error container	Patrick Jeremie Maniaul	1/25/2025	Ensure error messages reset upon successful password reset	Enki Prince	1/30/2025	Passed

ERR010	Successful login allows navigation back to the sign-in page without destroying session	Patrick Jeremie Maniaul	1/25/2025	Implement session handling to restrict access to the sign-in page after login	Enki Prince	1/30/2025	Passed
ERR011	Login attempts validation missing	Patrick Jeremie Maniaul	1/25/2025	Implement login attempt tracking and lockout mechanisms	Enki Prince	1/30/2025	Passed